

MATHEMATICS GRADE 10: STATISTICS I

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 4.1 (9 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Mathematics
Strand	Strand 4.0: Statistics and Probability
Sub-Strand	Sub-Strand 4.1: Statistics I
Total Duration	9 lessons × 40 minutes = 360 minutes total
Sub-Strand Content	– Frequency distribution tables (ungrouped and grouped data) – Measures of central tendency: mean, mode and median (ungrouped and grouped data) – Representation of data: histograms, frequency polygons and ogives – Interpretation of data from tables and graphs – Importance of collecting, organizing and interpreting data in making effective decisions
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) organise data using frequency distribution tables b) calculate the mean, mode and median of ungrouped and grouped data c) represent data using appropriate statistical graphs d) interpret data presented in tables and graphs e) appreciate the importance of collecting, organising and interpreting data in making effective decisions
Core Competencies	– Critical thinking and problem solving: learners compare two sets of marks, analyse the evidence and justify predictions about the next steps for learners and teachers. – Communication and collaboration: learners discuss and defend their interpretations of the data in groups and present findings to the class. – Digital literacy: learners may use calculators or spreadsheets to organise and graph data. – Learning to learn: learners revise their statistical model of the data as new measures and representations are introduced.
Core Values	– Integrity: learners record and report data honestly without altering scores. – Responsibility: learners use data to make fair and informed decisions about learning. – Unity: learners work cooperatively in groups to analyse and interpret data.
Science & Engineering Practices	– Analyzing and interpreting data: learners order scores, build frequency tables, compute averages and interpret graphs. – Developing and using models: learners build and revise a statistical model (table, measures, graph) of the two subjects' performance. – Using mathematics and computational thinking: learners calculate mean, median and mode and use them to compare data sets.

	– Engaging in argument from evidence: learners justify predictions about next steps using statistical evidence. – Asking questions and defining problems: learners add questions to the Driving Question Board.
Pertinent & Contemporary Issues (PCIs)	– Education for sustainable development / decision making: learners use data to make evidence-based decisions. – Life skills (critical thinking, decision making): learners interpret results to choose appropriate actions. – Citizenship: learners understand how data supports fair planning in school and community.
Career Connections	– Statistician: collects and analyses data to inform decisions, for example at the Kenya National Bureau of Statistics. – Teacher: uses class results to plan remedial teaching. – Data analyst: summarises and visualises data for organisations in Nairobi. – Agricultural extension officer: analyses crop yield data from farms in the Rift Valley. – Economist or business manager: uses data to plan and forecast.
Focus for Lessons	This unit is anchored in a comparison of 20 learners' scores in two subjects, where one set is high and the other is low. Learners organise, summarise, represent and interpret the data, building a statistical model across the lessons. They use it to predict what learners and teachers should do next.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can you use the two sets of data to predict the learners' and the teachers' next steps? KICD KEY INQUIRY QUESTIONS: 1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Anchoring Phenomenon	Learners are given the test scores of the same 20 students in two subjects. Subject 1: 72, 85, 90, 67, 78, 81, 95, 88, 74, 69, 82, 76, 91, 84, 77, 70, 89, 73, 80, 86. Subject 2: 35, 42, 27, 58, 44, 31, 60, 29, 15, 30, 32, 51, 60, 11, 13, 32, 23, 18, 21, 45. They arrange each set of scores in ascending order and discuss whether the learners passed or failed. They compare the two subjects and use their findings to predict the next steps for the learners and the two subject teachers. It is puzzling because the same students scored very high in one subject and very low in the other. The raw lists make it hard to see patterns, spread or what action is needed without organising and summarising the data.
Storyline Thread	Lesson 1: Phenomenon – learners predict, order the two sets of scores in ascending order, discuss pass or fail, and raise questions for the DQB. Lesson 2: Frequency distribution tables for ungrouped data – tally and organise the scores. Lesson 3: Grouped frequency distribution tables – choose class intervals and classify the scores of both subjects. Lesson 4: Mean, mode and median of ungrouped data – compare the two subjects. Lesson 5: Mean, mode and median of grouped data – estimate the measures from the grouped tables. Lesson 6: Representation of data I – histograms and frequency polygons of the two subjects. Lesson 7: Representation of data II – cumulative frequency and ogives; estimate the median and quartiles. Lesson 8: Interpretation of data – read tables and graphs to compare the subjects and predict next steps for learners and teachers. Lesson 9: Importance of collecting, organising and interpreting data in making effective decisions – final model revision and recommendations answering the driving question.
Supporting Phenomena	– Two Kenyan football teams in the Kisumu league with the same number of goals but different consistency across matches. – A farmer in Kericho recording daily tea leaf yields and needing a typical value to plan. – A school canteen in Nairobi recording daily sales of mandazi to decide how many to prepare.

	– Daily maximum temperatures in Mombasa and Nakuru compared to decide what to expect.
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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LESSON 1 (80 minutes (2 x 40-minute periods)): The Two Sets of Scores: Phenomenon Launch

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To launch the anchoring phenomenon: the same 20 learners in a Kisumu day secondary school scored very high in Subject 1 and very low in Subject 2. Learners predict, arrange both lists in ascending order, decide who passed or failed, and raise questions showing that raw lists are hard to read without organising and summarising data.
Knowledge	Raw data can be arranged in ascending order to see the highest score, lowest score and pass/fail at a glance. With a pass mark of 50 out of 100, all 20 learners pass Subject 1 (lowest 67, highest 95, range 28) while only 4 pass Subject 2 (51, 58, 60, 60) and 16 fail (lowest 11, highest 60, range 49). The raw lists still raise questions about patterns, spread and what action is needed.
Skills	Arranging data in ascending order; reading the highest, lowest and range from ordered data; counting passes and fails against a pass mark; making and justifying predictions from evidence; asking investigable questions (NGSS practice: Asking Questions and Defining Problems; Analyzing and Interpreting Data).
Attitudes	Curiosity about what data can reveal; honesty and accuracy when recording scores; respect for learners and teachers by treating reasons for scores as hypotheses and not blame; cooperation in groups of 4.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	This is the launch lesson. It creates the need to know: the raw lists hide patterns, spread and the action needed. The questions on the Driving Question Board (DQB) drive Lessons 2 to 9, where learners will build frequency tables, calculate measures, draw graphs and finally write recommendations for Madam Achieng.
Safety Notes	No hazardous materials. Seat groups so that no learner's marks are singled out publicly; the scores are labelled Learner 1 to Learner 20 only. Remind learners that the scores describe test results, not a person's worth, and that we do not blame any teacher or learner.

B. LESSON OVERVIEW

Madam Achieng, the Deputy Principal of a day secondary school in Kisumu near Lake Victoria, gives the Grade 10 class the test scores of the same 20 learners in two subjects, and asks them to use the data to decide next steps. Subject 1 is taught by Mr. Otieno and Subject 2 by Mrs. Wanjiku. Learners first predict what they expect, then work in groups of 4 to arrange both lists in ascending order, identify highest and lowest scores, and use the class pass mark of 50 to count passes and fails (20 pass Subject 1; 4 pass and 16 fail Subject 2). They notice the puzzle: the same learners did very well in one subject and poorly in the other. Because the ordered lists still do not show patterns or spread clearly, learners raise questions and open the Driving Question Board with the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps? They build an initial model of what they think is happening and what they need to find out.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners sit in groups of 4. They are told: 'Madam Achieng, the	Say: 'Imagine you are Madam Achieng in the staffroom in Kisumu	Predict-Observe-Explain (prediction first): committing to a	Observable indicators: (1) every learner has a written prediction

 VIDEO: Range and mid-range Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Stem-and-Leaf Plots, Range of a Data Set (Flexbook) Source: CK-12  Search ARES for similar readings	Deputy Principal, has the test scores of the same 20 Grade 10 learners in Subject 1 and Subject 2. Mr. Otieno teaches Subject 1 and Mrs. Wanjiku teaches Subject 2.' Before seeing any numbers, each learner writes in their notebook: (a) Do you expect the 20 learners to do equally well in both subjects? (b) What would you look for in the scores to decide who passed? (c) What do you think Madam Achieng will do next? They share predictions in their group and record one group prediction on manila paper. Then they are shown the two raw lists (in the order of Learner 1 to Learner 20): Subject 1: 72, 85, 90, 67, 78, 81, 95, 88, 74, 69, 82, 76, 91, 84, 77, 70, 89, 73, 80, 86. Subject 2: 35, 42, 27, 58, 44, 31, 60, 29, 15, 30, 32, 51, 60, 11, 13, 32, 23, 18, 21, 45. They look quickly at the lists and revise or confirm their prediction, writing one sentence on what they notice.	with two lists of marks. Before you touch a pen, what do you expect?' Give each learner 2 minutes of silent writing. Then say: 'Turn to your group and share. Everyone must speak.' Circulate and listen for predictions such as 'same learners so same marks'. Display the raw lists and say: 'Look for 30 seconds without writing. What do you notice first?' WAIT TIME 10-15 seconds after asking, then cold-call 4 learners from different groups. Ask: 'Which list looks higher? How can you tell?' and 'Does the order of the numbers help you?' Record the learners' words on the board under the heading 'What we notice' without correcting them. Ask one learner: 'Learner 7 scored 95 in Subject 1 and 60 in Subject 2. What does that tell us, and what does it not tell us?' Do not give reasons for the difference.	written prediction before seeing the data exposes learners' prior ideas (for example that a learner who is strong in one subject is strong in all) so that the later evidence can confirm or challenge them. Think-Pair-Share builds confidence before whole-class talk.	with a reason in their notebook before the data are shown; (2) at least 4 of 5 groups record a group prediction on manila paper; (3) learners can state at least one observation that compares the lists, such as 'Subject 1 numbers look bigger'; (4) learners avoid blaming statements and use words like 'may' or 'could'. Note learners who assume the same learners must score the same in both subjects, for follow-up.
Observe Phase  VIDEO: Range and mid-range Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Stem-and-Leaf Plots, Range of a Data Set (Flexbook) Source: CK-12  Search ARES for similar readings	Each group receives a printed copy (or a copy written on the board) of both raw lists. Step 1: Each learner in the group arranges ONE list in ascending order (two learners per subject) and the pairs then swap and check each other's work. The correct ordered lists are: Subject 1: 67, 69, 70, 72, 73, 74, 76, 77, 78, 80, 81, 82, 84, 85, 86, 88, 89, 90, 91, 95. Subject 2: 11, 13, 15, 18, 21, 23, 27, 29, 30, 31, 32, 32, 35, 42, 44, 45, 51, 58, 60, 60. They check that each list still has 20 scores (n = 20). Step 2: They record the lowest score, highest score and range (highest minus lowest) for each subject: Subject 1 lowest 67, highest 95, range 28; Subject 2 lowest 11,	Say: 'Each pair takes one list. Cross out each score on the raw list as you copy it so that none is missed or repeated.' Circulate; ask 'How do you know you have all 20 scores?' Prompt a check: 'Count your ordered list. Is it 20?' Watch for common errors such as omitting a repeated 32 or 60 in Subject 2, or ordering 9 after 10. For groups that finish early ask: 'If the pass mark was 40, how many would pass Subject 2?' (7). Pause the class and say: 'Hands down. In your group, find the lowest in Subject 1 and the highest in Subject 2. What do you notice?' (67 is higher than 60.) WAIT TIME 10-15 seconds, then cold-call 3 groups to report the range for each	Structured data organisation with a claim-evidence check: ordering the scores converts an unreadable list into one where highest, lowest and pass/fail are visible at a glance, and learners use the line at 50 as evidence for their claims. Cross-checking with a partner builds accuracy and a habit of verifying data (NGSS: Analyzing and Interpreting Data).	Observable indicators: (1) both ordered lists are correct and contain 20 scores, including two 32s and two 60s in Subject 2; (2) learners correctly state lowest, highest and range (67, 95, 28; 11, 60, 49); (3) learners state 20 pass in Subject 1 and 4 pass and 16 fail in Subject 2 using the pass mark of 50; (4) learners can say what the ordered lists still do not show (for example how scores cluster or spread). Use a quick tally of groups with correct counts; reteach ordering with a smaller sample (for example 5 scores) for any group that gets it wrong.

	highest 60, range 49. Step 3: Using the class pass mark of 50 out of 100, they draw a line on each ordered list and count passes and fails: Subject 1, 20 pass and 0 fail; Subject 2, 4 pass (51, 58, 60, 60) and 16 fail. They note on manila paper that 7 learners in Subject 2 scored 40 or more, and record the table 'Subject Lowest Highest Range Pass Fail'. Step 4: Groups write what they notice about the two subjects and what they still cannot tell from the lists.	subject and 2 groups to report passes and fails. Say: 'Let us record our results in one class table on the board.' Ask: 'Can you tell from these ordered lists whether most scores are near the middle, or spread out evenly?' Let learners struggle and say this is difficult. Do not compute the mean or median yet.		
Explain Phase  VIDEO: Statistics intro: Mean, median, & mode <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Stem and Leaf Plots, Range of a Data Set Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners hold a whole-class discussion using the evidence from their tables. They answer in their groups: (1) Did the learners pass or fail Subject 1? Subject 2? Use the numbers. (2) Can we say the 20 learners are 'good' or 'weak'? Why or why not? (3) What do you think Madam Achieng should do for Mr. Otieno's class and for Mrs. Wanjiku's class? (4) What might explain the difference? Each group writes at least two possible reasons as hypotheses (for example 'the test may have been harder', 'the topics may not have been revised', 'the teaching method could differ') and one piece of data we would need to test each. They then complete a sentence frame: 'The evidence shows that ____, but we cannot yet say ____ because ____.' Groups present to the class from their manila paper.	Say: 'Use your numbers: who passed, and how do you know?' Then: 'The same 20 learners scored between 67 and 95 in Subject 1 and between 11 and 60 in Subject 2. Can we say these learners are weak? Turn and talk for 1 minute.' WAIT TIME 10-15 seconds, then cold-call 3 learners. Press for evidence: 'Which number supports your claim?' When a learner offers a cause as a fact (for example 'Mrs. Wanjiku is not teaching well'), respond: 'That is a hypothesis, not yet a fact. What data would we need to test it?' Write 'Hypothesis' and 'Fact from the data' in two columns on the board and sort the learners' statements with the class. Facts: all 20 pass Subject 1; 16 fail Subject 2. Hypotheses: harder test, less revision, teaching method. Ask: 'What decisions could Madam Achieng make today, and what decisions would be unfair without more data?' Invite 2 groups to present their sentence frames. Use local analogies if needed: 'If one farmer in the Rift Valley gets a poor maize yield from one plot, do we blame the farmer or check the soil, rain and	Claim-Evidence-Reasoning with a Fact vs Hypothesis sort: learners separate what the data show from what is only suggested, building scientific habits of mind and fairness. The sentence frame forces them to state the limits of the evidence.	Observable indicators: (1) learners use specific numbers (20 pass; 4 pass, 16 fail; lowest 67 and 11) to support claims; (2) at least 3 groups place causes under 'Hypothesis' and not 'Fact'; (3) learners recognise that the difference is between subjects, not that all learners are weak; (4) each group can name data needed to test one hypothesis (for example a second test or topic-by-topic marks). Listen for blame language and redirect it to evidence.

		seed first?'		
Driving Question Board (DQB) Creation  VIDEO: Interquartile range (IQR) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Median & range puzzlers <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners read the driving question aloud: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' Each learner writes 2 questions on sticky notes or paper strips about what they want to find out, thinking about patterns, spread, typical scores and actions needed. Groups sort their notes into categories such as 'How to organise the data', 'How to find a typical score', 'How to compare spread', 'How to show it in graphs' and 'What decisions to make'. Each group posts its 3 best questions on the DQB. Possible learner questions: 'How many learners got each score?', 'What is the average score in each subject?', 'Is there a faster way to see where most scores lie?', 'How can we show the two subjects in a picture?', 'Which learners need help?', 'What more data do we need?' The class then links the board to the two KICD inquiry questions: How do we organise data using frequency distribution tables? Why is it important to collect, organise and interpret data in making decisions?	Say: 'The ordered lists helped us, but they did not tell us everything. What are you still wondering? Write 2 questions: one about how to organise or summarise the scores, and one about what action to take.' Give 3 minutes of silent writing. Circulate and push for investigable questions: 'Can we answer that using the data? What would we calculate or draw?' Convert yes/no questions: 'Is Subject 2 hard?' becomes 'How can we compare the spread of scores in Subject 2 with Subject 1?' Ask each group to post 3 questions and cold-call 5 groups to read one aloud and say where it fits. WAIT TIME 10-15 seconds before asking the class: 'Which questions look similar? Which can we answer first?' Group them with the class into 4-5 themes and label each with a lesson cluster (frequency tables, measures of the typical score, graphs, comparing, decisions) without giving answers. Add the two KICD key inquiry questions to the board on strips. Say: 'We will return to this board at the end of each lesson to move questions to answered.'	Driving Question Board with question sorting: learner-generated questions give ownership of the unit, make the need for frequency tables, measures and graphs visible, and allow the class to track growing understanding across Lessons 2 to 9.	Observable indicators: (1) every learner contributes at least 2 questions; (2) at least half of the posted questions can be answered with data (for example about frequency, average, spread, graphs); (3) groups can sort questions into themes; (4) at least one question asks what more data is needed. Keep the notes to identify misconceptions, such as using only the highest score to judge a class.
Model Building Phase  VIDEO: Measures of spread: range, variance & standard deviation <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English	On manila paper, each group draws an initial model titled 'What we think is happening, and what we need to find out'. The model includes: (1) a simple box showing the two ordered lists with lowest, highest, range and pass/fail counts; (2) arrows linking 'Subject 1: all 20 pass, scores 67 to 95' and 'Subject 2: 4 pass, 16 fail, scores 11 to 60' to Madam Achieng's decisions; (3) a section 'Possible reasons (hypotheses)' with 'may'	Say: 'Draw your first model. It will be revised many times, so it is fine if it is incomplete. Include what the data show, what you guess, and what you need.' Circulate with prompts: 'Where did you use the pass mark?', 'Which parts of your model are facts and which are hypotheses?', 'What would you add if you could do one more calculation?' Set the gallery walk for 6 minutes and ask each learner to leave one question or	Initial model building with gallery walk feedback: externalising ideas in a diagram makes thinking visible, separates evidence from guesses, and creates a baseline model that learners will revise in each later lesson as they gain new tools (Developing and Using Models).	Observable indicators: (1) models include correct pass/fail counts (20 pass; 4 pass, 16 fail) and the correct lowest, highest and range; (2) hypotheses are labelled as possible and not as facts; (3) models name at least one tool or data still needed; (4) exit tickets show that learners can say what ordering revealed (highest, lowest, pass/fail) and what it did not (patterns, spread, typical score, action). Learners whose exit ticket

- US curriculum) Search ARES for similar videos EXERCISE: Interquartile range (IQR) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	or 'could' wording; (4) a section 'Tools we think we need' (for example tables, a typical score, pictures or graphs); (5) a section 'Next steps we predict' for the learners, Mr. Otieno and Mrs. Wanjiku. Groups do a gallery walk, adding one question or suggestion on a sticky note to another group's model. Finally each learner completes an exit ticket: 'One thing the ordered lists showed me is _____. One thing they did not show me is _____.'	suggestion on another group's model. WAIT TIME 10-15 seconds after asking: 'What do the models have in common?' Cold-call 3 learners. Say: 'Keep your model; in Lesson 2 we start organising the scores with tally marks and frequency tables.' Collect exit tickets. Photograph or keep the models for revision in later lessons. Close with Madam Achieng's request: 'Use the data to decide next steps. We are not ready to decide yet. What do we need?'	shows no limit of ordered data get extra support in Lesson 2.
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D. TEACHER REFLECTION

Did every learner correctly order both lists, including the repeated 32 and 60 in Subject 2? Which groups used the pass mark of 50 accurately? Did learners treat causes as hypotheses or did blame statements about Mr. Otieno, Mrs. Wanjiku or the learners occur, and how did I redirect them? Which DQB questions will be answered in Lessons 2 to 9, and which need extra data collection? What prior ideas about marks, averages or pass/fail did I notice that need to be revisited?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the same 20 learners scored between 67 and 95 in Subject 1 (range 28, all 20 pass) and between 11 and 60 in Subject 2 (range 49, only 4 pass and 16 fail at a pass mark of 50). Ordering the scores made the highest, lowest and pass/fail easy to see.
What did I learn?	We learned that arranging raw data in ascending order shows the highest, lowest and pass/fail at a glance, but it does not clearly show patterns, spread or the typical score. Reasons for the difference are only hypotheses until we have more data.
How does this explain the phenomenon?	The difference is between the subjects, so we cannot say all 20 learners are weak. To decide next steps for the learners, Mr. Otieno and Mrs. Wanjiku, we need to organise and summarise the data with tables, typical values and graphs, which is what we will do in the coming lessons.

LESSON 2 (40 minutes (one lesson; extend to 80 minutes if a double lesson is available)): Frequency Distribution Tables for Ungrouped Data: Tallying the Two Sets of Scores

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners build ungrouped frequency distribution tables (score, tally, frequency) for the two sets of 20 scores using tally marks grouped in fives, and check that the frequencies add up to $n = 20$. They then judge how much the tables help to answer the driving question and what the tables still cannot show.
Knowledge	A frequency distribution table lists each score with a tally and the number of times it occurs (its frequency, f). The frequencies must add up to the total number of scores, $n = 20$. In Subject 1 every one of the 20 scores occurs exactly once ($f = 1$ for each). In Subject 2, 32 and 60 each occur twice and the other 16 scores occur once, which gives 18 different scores and a total of 20.
Skills	Organising data using tally marks grouped in fives; constructing an ungrouped frequency table from the ordered lists; checking totals (sum of $f = n$); identifying the most frequent scores; analysing and interpreting data (NGSS SEP 4); using mathematics and computational thinking (NGSS SEP 5) to check that the frequencies add to 20; constructing explanations from evidence (NGSS SEP 6).
Attitudes	Accuracy and patience when tallying; honesty in checking and correcting counts; cooperation in groups of 4; curiosity about what the data can and cannot tell us; willingness to say 'we need more data' instead of guessing reasons.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	In Lesson 1, learners put both raw lists in ascending order and saw the highest, lowest and pass/fail counts, but found the raw lists still raise questions about patterns, spread and action needed. Lesson 2 gives learners their first organising tool, the ungrouped frequency table. The tables show that Subject 1 has every score appearing once and Subject 2 has two repeated scores (32 and 60). This creates the need for grouping in Lesson 3 (not taught here) and the need for summary measures in Lesson 4.
Safety Notes	No hazardous materials. Use a pencil and ruler on manila paper or exercise books. Learners should remain seated while tallying to avoid accidents with sharp markers or tape. Be sensitive about the scores: refer to Learner 1 to Learner 20 by number, do not shame learners for low scores, and treat the Subject 2 scores as data about the subject, not as a judgement of the learners.

B. LESSON OVERVIEW

Madam Achieng, the Head of Department at a day secondary school in Kisumu, still has the same 20 Grade 10 learners' scores in Subject 1 (Mr. Otieno's class) and Subject 2 (Mrs. Wanjiku's class). Last lesson the class ordered the scores and saw that all 20 learners passed Subject 1 (lowest 67) while only 4 of 20 passed Subject 2 (lowest 11), using the pass mark of 50. The ordered lists are still long, so Madam Achieng asks: 'Can we organise the scores so that a busy teacher can see how often each score occurs?' Learners first predict what a table of the scores might look like and how many times each score will appear. They then build ungrouped frequency tables with tally marks grouped in fives, one table for each subject, and check that the frequencies add up to 20. In the Explain Phase they compare the two tables. Subject 1 has 20 different scores, each with $f = 1$, so the table is as long as the list and shows little. Subject 2 has 18 different scores with 32 and 60 each occurring twice. Learners discuss what the tables help them see and what questions remain about spread and action needed, and they treat reasons for the difference as hypotheses, not facts. They add new questions to the Driving Question Board and add the frequency table to the class model of how data helps predict next steps for the learners and teachers.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Frequency tables & dot plots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Creating frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall Lesson 1: both ordered lists and the pass/fail counts (20 pass in Subject 1; 4 pass and 16 fail in Subject 2). Each learner receives the ascending lists. Subject 1: 67, 69, 70, 72, 73, 74, 76, 77, 78, 80, 81, 82, 84, 85, 86, 88, 89, 90, 91, 95. Subject 2: 11, 13, 15, 18, 21, 23, 27, 29, 30, 31, 32, 32, 35, 42, 44, 45, 51, 58, 60, 60. Learners individually write answers to: (a) 'If we make a table showing each score and how many learners got it, how many rows might Subject 1 have? Subject 2?' (b) 'Will any score appear more than once? Which subject is more likely to have repeats?' (c) 'Will a table help Madam Achieng decide the next steps?' They share predictions with a partner, then record them on a 'Predictions' column on the board and in their books.	Display the Driving Question: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' Say: 'Yesterday we ordered the scores. Madam Achieng says the lists are still long. How can we show how often each score occurs?' Give 2 minutes of silent individual writing, then WAIT TIME of 10-15 seconds after each question before taking responses. Cold-call 4 learners for predicted row counts and 3 learners for predictions about repeats; ask each 'What in the ordered list makes you say that?' Do not confirm or correct yet. Record all predictions on the board under 'Predictions' with the learner's reasoning. Say: 'Hold on to these. We will test them with our own tally tables.'	Predict-Observe-Explain (POE) with Think-Pair-Share: committing to a written prediction before tallying creates a cognitive conflict or confirmation that makes learners care about the result. Pair sharing lets learners hear other reasoning before they test their own ideas against the evidence.	Observe whether learners (1) write a numerical prediction for the number of rows, (2) give a reason that refers to the ordered list (for example 'the Subject 2 list has 32 twice and 60 twice'), and (3) distinguish repeats from no repeats. Note learners who predict that Subject 1 will have the same number of rows as scores (an accurate prediction) and those who do not look at the lists. Tally common predictions on the board to plan the Explain Phase.
Observe Phase  VIDEO: Frequency tables & dot plots Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reading dot plots & frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners build the ungrouped frequency distribution tables. Step 1: Teacher models the tally process using a few scores from Subject 2 (for example the score 32: strike the first tally, then the second), showing that the fifth tally crosses the first four so tallies are grouped in fives. Step 2: Each group draws two tables on manila paper with three columns: Score (x), Tally, Frequency (f). Step 3: Using the ORIGINAL RAW lists as in the order given (Subject 1: 72, 85, 90, 67, 78, 81, 95, 88, 74, 69, 82, 76, 91, 84, 77, 70, 89, 73, 80, 86; Subject 2: 35, 42, 27, 58, 44, 31, 60, 29, 15, 30, 32, 51, 60, 11, 13, 32, 23, 18, 21, 45), one learner	Model for 3 minutes on the board, then circulate. Say: 'Read each score aloud, make one tally, and cross it off the raw list so nothing is counted twice.' Ask groups: 'Where did you put the fifth tally?' and 'How do you know you have not missed a score?' Intervene with questions, not answers, if a group's total is not 20: 'Your frequencies add to 19. Where might a score be missing? How can you find it?' Give WAIT TIME of 10-15 seconds before offering a hint. Tell groups to compare with the ordered list from Lesson 1 to check for repeats. Cold-call 3 groups to report total frequency for Subject 1 and 3 groups for Subject	Structured data collection with role rotation (reader, tallier, checker, recorder) and self-checking against the known total $n = 20$. Learners build understanding that a table is a systematic count: each score is counted once, and a check total shows whether the count is accurate (using mathematics and computational thinking, NGSS SEP 5).	Look for: (1) tallies grouped in fives with the fifth tally crossing the first four; (2) a row for every score with tally and frequency in the correct column; (3) frequency columns adding to 20 in both tables; (4) correct identification that Subject 1 has no repeats and Subject 2 repeats 32 and 60; (5) groups that find and correct their own counting errors. Note groups that count tally marks wrongly (for example counting a group of five as 4) and address this in the Explain Phase.

	reads out the scores one by one, a second tallies in the table, a third crosses off each score in the raw list, and the fourth checks. The group then swaps roles for the second subject. Step 4: For each table, groups write the frequency as a number next to the tally and add the frequency column. They must get 20 for each table. Step 5: Groups record how many different scores each subject has (20 in Subject 1; 18 in Subject 2) and which scores have a frequency greater than 1 (none in Subject 1; 32 and 60 in Subject 2). Groups that finish early identify which row of the Subject 2 table includes Learner 7's score (60, which has $f = 2$) and which score in Subject 1 belongs to Learner 7 (95, $f = 1$).	2 and record on the board. Expected: Subject 1 total 20, all $f = 1$; Subject 2 total 20, with 32 and 60 each $f = 2$. Do not tell learners why scores differ between subjects; if asked, say 'Good question. The table cannot tell us why. We will write that as a question for the Driving Question Board.'		
Explain Phase  VIDEO: Ungrouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Creating frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners use their finished tables to answer questions in a whole-class discussion and then in a short written response. Discussion questions: (1) 'Did your predictions match your tables? Which prediction was correct and which changed?' (2) 'Subject 1 has 20 different scores, each with $f = 1$. What does this table show that the ordered list did not? What does it not show?' (3) 'Subject 2 has 32 and 60 each with $f = 2$. How does the table show this?' (4) 'Why must the frequencies add up to 20? What does it mean if they add to 19 or 21?' (5) 'Madam Achieng wants to see quickly where most learners scored. Can she do that from the Subject 1 table? From the Subject 2 table?' Learners should explain that the ungrouped tables are accurate and show the repeats, but they are long and, for Subject 1, are almost as long as the list; the tables do not yet show	Ask the questions one at a time with WAIT TIME of 10-15 seconds before each response. Use a Talk Move: 'Who can add on to what Learner 5 said?' Cold-call 5 learners in total across the questions, including at least 2 who did not speak in the Observe Phase. Press for evidence: 'Point to the row in your table that shows that.' When a learner says 'Subject 1 is easy and Subject 2 is hard', respond: 'That could be one explanation. Is it in our data? What data would we need to test it?' and write the hypothesis on a 'Hypotheses to test' chart. Revisit the Predict Phase predictions on the board: tick those supported by evidence and amend others with the learners' words. Emphasise: 'The total frequency is 20 because there are 20 learners. A table that does not add to 20 has a counting error.' Close by saying: 'Our tables are accurate, but they are still	Claim-Evidence-Reasoning (CER) with a whole-class Talk Move discussion (NGSS SEP 6, constructing explanations): learners must link their statements to specific rows in their tables. Treating reasons for the difference as hypotheses to test teaches the difference between what data show and what we infer.	Check that written CER statements: (1) cite specific frequencies (for example '$f = 2$ for 32 and 60'); (2) correctly state that sum of $f = 20$ for both tables; (3) acknowledge a limitation of the ungrouped table (long, no clear clusters for Subject 1); (4) use cautious language ('may', 'could', 'we need more data') for causes. Look for misconceptions: thinking that frequency means the score itself, or that the table shows who scored what (it does not identify individual learners). Collect 5 written responses as an exit check.

	clusters or spread clearly, so another way of organising is needed (to be developed in Lesson 3). Learners then complete a claim-evidence-reasoning statement: 'Claim: _____. Evidence from the table: _____. Reasoning: _____. For example: 'Claim: the ungrouped table for Subject 1 does not make the pattern clearer. Evidence: all 20 scores have $f = 1$ and the table has 20 rows. Reasoning: a table that is as long as the list does not summarise the data.' Learners also write one honest statement about causes, such as 'The table cannot tell us why Subject 2 scores are lower; reasons like the test difficulty, teaching method or revision time could be possible, but we need more data.'	long. What would help Madam Achieng see where most scores lie? Keep that question for the next lesson.'		
Driving Question Board (DQB) Creation  VIDEO: Statistics intro: Mean, median, & mode <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Reading dot plots & frequency tables Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners return to the Driving Question Board with the question 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' Individually, each learner writes one thing the frequency tables helped them see (for example 'Subject 2 has two repeated scores, 32 and 60') and one new question on a sticky note or paper strip (for example 'How can we show where most Subject 2 scores lie without so many rows?', 'Why do Subject 2 scores spread from 11 to 60?', 'What is the typical score in each subject?', 'How do we make the Subject 1 table shorter?'). They post notes on the DQB under headings: 'What the tables showed', 'Questions we can answer next' and 'Questions that need more data' (for example 'Why did learners score differently in the two subjects?'). The class sorts the	Say: 'Our DQB should show what we know and what we still need. Write what the tables helped you see and one new question.' Give 2 minutes of silent writing, then WAIT TIME of 10-15 seconds before sharing. Cold-call 4 learners to read and post a question and ask the class 'Which heading does this belong under, and why?' Highlight questions that cannot be answered from our data (causes) and place them under 'Questions that need more data'. Group similar questions and point out those the class will return to: 'The question about showing where most scores lie is exactly what we will tackle when we group scores into classes.' Do not promise answers to questions that require extra data collection, but say 'We will decide what data we would need to collect.'	Driving Question Board with question sorting: making learners' questions visible and classifying them as answerable now, answerable later in the unit, or needing more data builds ownership of the inquiry and teaches that statistical questions depend on the data available (NGSS SEP 1, asking questions).	Check that notes (1) state a specific observation from the tables, (2) include a genuine question connected to the driving question, and (3) are placed under a sensible heading. Look for questions about organising (grouping, averages, graphs), which show progress toward Key Inquiry Question 1, and questions about decisions, which connect to Key Inquiry Question 2. Count how many learners ask about causes and address the need for more data.

	questions together and links two or three to future lessons (grouping, averages, graphs).			
Model Building Phase  VIDEO: Two-way relative frequency tables <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Ungrouped Data to Find the Mean (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners add Lesson 2 to the class model of 'How data helps us predict next steps'. The model on the wall is a flow diagram: Raw lists → Ordered lists (Lesson 1) → Frequency tables (Lesson 2) → [Grouped tables, measures, graphs to come] → Interpretation → Next steps for learners and teachers. In groups of 4, learners draw a mini version in their notebooks and add one box or arrow for the frequency table describing what it does (counts how often each score occurs) and what it still cannot do (does not clearly show clusters and spread for a large range of different scores). They write a one-sentence tentative prediction for the next steps for Mr. Otieno and Mrs. Wanjiku based on evidence so far (for example 'Subject 1: all 20 pass; Subject 2: only 4 of 20 pass, so Subject 2 may need extra support, but we need more information'). They complete the Summary Table: What we observed, What we learned, How we can explain it.	Display the class model and say: 'Last lesson we added ordered lists. What do we add today, and what does it help with?' Give WAIT TIME of 10-15 seconds, then cold-call 3 groups to suggest an addition to the model. Probe: 'What can the frequency table do that the ordered list could not? What can it still not do?' Correct only factual errors (for example if a group says Subject 1 has a mode: 'Check your table. How many scores have f greater than 1?'). Remind learners to treat recommendations as tentative: 'Based on the data so far, what can we say, and what do we still need to find out?' Record the agreed addition on the class model and distribute the Summary Table. Collect the notebooks and tables for review. Optional homework: use the ungrouped table to tally 20 daily mandazi sales for a stall in Mombasa (teacher-provided list) and check that the frequencies add to 20.	Cumulative model revision with a Summary Table (Observed-Learned-Explained): learners revise a visible model each lesson, connecting new evidence to earlier evidence and to the driving question, so understanding of how data supports decisions grows in steps (NGSS SEP 2, developing and using models).	Check that each group's model box (1) states what a frequency table does, (2) names a limitation, and (3) links to the driving question with a cautious prediction. Check that the Summary Table includes accurate details: both tables total 20; Subject 1 all f = 1; Subject 2 has 32 and 60 each with f = 2. Note groups that overstate causes and return to the 'Hypotheses to test' chart.

D. TEACHER REFLECTION

Which learners built accurate tables, and which needed help with tally grouping or with checking the total of 20? Did the Predict Phase predictions about the number of rows and repeats help learners care about the result, and which predictions changed? How well did learners distinguish what the tables show from why the scores differ? Did anyone claim a cause as a fact, and how did I respond? Did the observation that the Subject 1 table is as long as the list lead learners to the need for a shorter way of organising (Lesson 3)? What will I change about group roles, wait time or cold-calling next time?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We tallied the 20 scores in Subject 1 and the 20 scores in Subject 2 into frequency tables with columns Score, Tally and Frequency. In Subject 1 every score occurred once (20 different scores). In Subject 2, 32 and 60 each occurred twice and the other scores once (18 different scores). Both frequency columns added to 20.
What did I learn?	A frequency distribution table for ungrouped data lists each score, a tally (grouped in fives) and the frequency, and the frequencies must add up to $n = 20$. Tables help us count accurately and spot repeats, but ungrouped tables with many different scores are long and do not yet clearly show where scores cluster or how widely they spread.
How does this explain the phenomenon?	The tables are accurate summaries of how often each score occurs, which helps Madam Achieng, Mr. Otieno and Mrs. Wanjiku check the data. However, Subject 1's table is as long as its list, so we need another way of organising the scores to see patterns and spread. The tables cannot tell us why the subjects differ; that would need more data.

LESSON 3 (40 minutes (one double lesson of 80 minutes may be used if learners need more time to classify)): Grouped Frequency Distribution Tables: Choosing Class Intervals and Classifying the Scores

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners choose a suitable class width and class limits and classify the scores of Subject 1 and Subject 2 into grouped frequency distribution tables, so that patterns and spread that the raw and ungrouped lists hid become visible and can be used to predict next steps for the learners, Mr. Otieno and Mrs. Wanjiku.
Knowledge	Learners will be able to: (1) use the range (28 for Subject 1 and 49 for Subject 2) to decide on a sensible class width; (2) write class intervals with class limits (10-19, 20-29, ..., 90-99) and state class boundaries (9.5-19.5, ..., 89.5-99.5) and midpoints (14.5, 24.5, ..., 94.5); (3) classify data with tally marks and complete a grouped frequency table with frequencies adding to $n = 20$.
Skills	Planning and carrying out investigations (organising and classifying data); analysing and interpreting data (reading clusters and gaps in the grouped table); using mathematics and computational thinking (range, class width, tallying, checking totals); constructing explanations from evidence; communicating findings on manila paper; collaboration in groups of 4.
Attitudes	Learners show accuracy and patience when tallying, curiosity about what the grouped data says, respect for others' ideas when groups choose different class widths, and honesty by treating reasons for the score difference as hypotheses and not facts.
Key Inquiry Question	KICD Key Inquiry Questions: 1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions? Lesson focus: Why does grouping scores into class intervals help us see patterns that 20 separate scores do not?
Purpose in Storyline	In Lesson 2 learners built ungrouped tables with tally marks and saw that every Subject 1 score had frequency 1, and Subject 2 had only 32 and 60 occurring twice. That table was long and showed little. Lesson 3 answers the question 'how can we organise this better?' by grouping. The grouped table (with boundaries and midpoints) is the foundation for Lesson 5 (grouped mean, mode, median), Lesson 6 (histograms and polygons) and Lesson 7 (cumulative frequency and ogives). It also gives the first clear evidence for the driving question: Subject 1 scores cluster between 60 and 99, while Subject 2 scores spread from 10 to 69.
Safety Notes	Low-risk classroom activity. Ensure learners share rulers and manila paper without crowding, keep walkways clear during gallery walk, and use sensitive language: scores belong to Learner 1 to Learner 20 and are not used to label or shame any learner. Remind the class that nothing in the data tells us why scores differ.

B. LESSON OVERVIEW

Madam Achieng's question remains: what should the learners and the two teachers do next? Learners recall the Lesson 2 ungrouped tables, where Subject 1 had 20 rows each with frequency 1, and predict how many rows a better table should have. Groups use the ranges (28 and 49) to test different class widths (5, 10, 20) on the ascending lists, then agree on a common class width of 10 so both subjects can be compared. They write class limits (10-19 to 90-99), tally all 20 scores of each subject, complete the grouped frequency tables (Subject 1: 0, 0, 0, 0, 0, 2, 7, 8, 3; Subject 2: 4, 4, 5, 3, 2, 2, 0, 0, 0), check totals of 20, and learn that class boundaries (9.5, 19.5, ..., 99.5) and midpoints (14.5 to 94.5) are needed for later graphs. Learners describe where each subject clusters, add new questions to the Driving Question Board and revise their cumulative model with the grouped evidence, treating reasons for the difference as hypotheses to investigate.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners look at their Lesson 2 ungrouped tables. Individually (2 minutes) they write a prediction in their notebooks: 'If we group the scores into classes such as 60-69, 70-79, how many rows will the Subject 1 table have, and which classes will have the most learners? What will Subject 2 look like?' They sketch a quick guess of the shape (where the tally marks will pile up). Then they turn to a partner and compare predictions.	Display on the board: 'Subject 1: every score occurs once. Subject 2: only 32 and 60 occur twice.' Say: 'Madam Achieng does not want a table with 20 rows. How could we squeeze the 20 scores into a few rows and still keep the story?' Give WAIT TIME of 10-15 seconds before anyone speaks. Cold-call 3 learners (not volunteers) to share predictions and record them on a 'Predictions' chart. Ask: 'What do you think will happen to Subject 1's tally marks, and why?' Do not correct predictions yet; say 'We will test this with evidence today.'	Predict-Share-Revisit: committing to a written prediction before instruction makes prior thinking visible and creates a reason to examine the evidence. Comparing with a partner exposes different ideas about what 'grouping' means, which the class will test later in the Explain Phase.	Observe whether learners (a) link the all-frequency-1 problem to the need to group, (b) predict that Subject 1 clusters in high classes and Subject 2 is more spread out, (c) give a reason using the ascending lists. Look for the misconception that grouping 'loses' scores or that classes can overlap. Record on a checklist the names of 5 learners who predicted a number of rows.
Observe Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Part A (10 min): In groups of 4, learners use the ascending lists and the ranges (Subject 1: $95 - 67 = 28$; Subject 2: $60 - 11 = 49$). Each group tries a different class width on manila paper (width 5, 10 or 20), counts how many classes are needed, and records advantages and disadvantages. Part B (12 min): The class agrees on one common width of 10 for both subjects, so they can be compared. Groups write the class intervals 10-19, 20-29, ..., 90-99 and tally the scores of Subject 1 and Subject 2 from the ascending lists, ticking each score off. Subject 1 working: 60-69 has 67, 69; 70-79 has 70, 72, 73, 74, 76, 77, 78; 80-89 has 80, 81, 82, 84, 85, 86, 88, 89; 90-99 has 90, 91, 95. Subject 2 working: 10-19: 11,	Introduce the idea: 'The range tells us how far the scores stretch. A good class width gives about 5 to 10 classes.' Circulate and ask: 'How many classes would width 5 give for Subject 2? Is that easier to read than 20 rows?' ($49 \div 5$ gives about 10 classes; $49 \div 10$ gives about 5 to 9 classes.) Prompt: 'Why 10-19 and then 20-29, not 10-20 and 20-30? Where would a score of 20 go?' Use WAIT TIME of 10-15 seconds after each question. Cold-call 4 groups to report their class widths and number of classes. Guide the whole class to agree on width 10 and the same classes for both subjects 'so we can compare Mr. Otieno's class and Mrs. Wanjiku's class fairly'. Check that classes do not overlap and every score fits in	Structured Data Collection and Classification (Planning and Carrying Out Investigations): learners compare class widths with evidence, which builds understanding of why class width is a choice that balances detail and readability. Using the same classes for both subjects builds the idea of a fair comparison.	Observable indicators: classes written without gaps or overlaps (10-19, 20-29, ...); tally marks grouped in fives; each score counted once only; frequencies total 20 for both subjects; Subject 1 has zeros for 10-19 to 50-59 and Subject 2 has zeros for 70-79 to 90-99. Common errors: placing 60 in 50-59, counting 32 only once, using classes of unequal width, or totals not equal to 20. Note groups needing help and ask them to recount using the working lists.

	13, 15, 18; 20-29: 21, 23, 27, 29; 30-39: 30, 31, 32, 32, 35; 40-49: 42, 44, 45; 50-59: 51, 58; 60-69: 60, 60. They complete the frequency columns, remembering tally marks in fives, and check that each total is 20.	exactly one class. Tell learners to tick each score off the list. Once totals are 20, ask: 'What would it mean if your total was 19 or 21?'		
Explain Phase  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar videos  HTML: Summary Statistics. Summarizing Univariate Distributions (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners complete the full grouped frequency table for both subjects side by side: class, tally, frequency (Subject 1: 0, 0, 0, 0, 0, 2, 7, 8, 3; Subject 2: 4, 4, 5, 3, 2, 2, 0, 0, 0; totals 20 and 20). The teacher then introduces class limits, class boundaries and midpoints: for 10-19 the lower boundary is 9.5, the upper is 19.5 and the midpoint is $(10 + 19)/2 = 14.5$. Groups add boundary and midpoint columns (9.5-19.5 to 89.5-99.5; midpoints 14.5, 24.5, 34.5, 44.5, 54.5, 64.5, 74.5, 84.5, 94.5). Each group writes 3 sentences on manila paper: (1) where Subject 1 scores cluster, (2) where Subject 2 scores cluster and spread, (3) what the grouped table tells Madam Achieng that the raw list did not. They use the language 'may', 'could' and 'we need more data' when suggesting why the subjects differ. Gallery walk: groups read other posters and place a sticky note with 'I agree because...' or 'I wonder...'.	Say: 'Look at the two columns of frequencies. Turn and talk: what do you notice?' WAIT TIME 10-15 seconds, then cold-call 4 learners. Press for evidence: 'Which class has the most learners in Subject 1? In Subject 2? How many learners in Subject 2 are in classes below 50?' ($4 + 4 + 5 + 3 = 16$, matching the earlier pass/fail count of 16 below 50.) Ask: 'Why do we need boundaries like 9.5 and 19.5? What gap would a graph have between 19 and 20?' Explain that boundaries close the gap so classes touch, which will be needed for histograms in Lesson 6. Do not teach calculations of mean, median or mode here; say 'those come in Lesson 5'. If a learner suggests a reason such as 'Subject 2 is harder' or 'the teacher is poor', respond: 'That is a hypothesis. What data would we need to test it?' Close with: 'Does the grouped table change your prediction from the start?'	Claim-Evidence-Reasoning with Gallery Walk: learners state a claim about each subject's clustering, support it with frequencies from the table, and reason about what it means for next steps. Peer feedback during the gallery walk allows learners to test their reasoning against that of other groups and to separate evidence from speculation.	Look for: correct statement that Subject 1 scores are concentrated in 70-79, 80-89 and 90-99 (18 of 20 learners), while Subject 2 scores are spread across six classes from 10-19 to 60-69; correct computation of boundaries and midpoints for at least 3 classes; statements that avoid claiming causes. Exit ticket (2 minutes): 'Write the class, boundaries and midpoint for the score 45, and say how many learners are in that class for Subject 2.' (40-49; 39.5-49.5; midpoint 44.5; frequency 3.) Collect and sort into got it, nearly, not yet.
Driving Question Board (DQB) Creation  VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i>	Learners return to the Driving Question Board: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' Individually they write on sticky notes (a) one thing the grouped tables helped them understand, and (b) one new question, for example 'What is the average of each grouped table?', 'How can we draw this table as a	Say: 'What new thing did the grouped table let us see?' Give WAIT TIME of 10-15 seconds before taking responses, then cold-call 3 learners to read a note aloud and say where it should go. Link notes to future lessons: 'Averages and the middle score come in Lessons 4 and 5, pictures of the table in Lesson 6.' Highlight any question about causes: 'Can	Driving Question Board as a Tracking Tool: learners see their understanding grow and sort questions by whether they can be answered with the evidence so far. Posting their own questions gives ownership and shows the logic of the sequence, from organising to summarising to graphing to deciding.	Check that notes reflect understanding (for example, 'grouping makes the clusters visible') and that new questions are mathematical and testable. Count how many learners posted a question about summaries or graphs, which shows readiness for Lessons 4 to 7. Listen for learners who ask about reasons and encourage them to turn the

Source: CK-12 Search ARES for similar videos HTML: Grouped Data to Find the Mean (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	picture?', 'Why is Subject 2 spread so widely?' or 'Would a second test give the same result?'. Groups place notes under headings such as 'Organising data', 'Summarising data', 'Graphing data' and 'Need more data'. The class marks questions answered by this lesson (for example 'How do we organise 20 scores?') and those to be answered in later lessons.	this table tell us WHY? What would we need?' Keep a visible list of 'evidence so far': tables show clustering; Subject 1 has 20 of 20 above 60; Subject 2 has 16 below 50. Revisit the KICD question: 'Why is it important to organise data in making decisions?' and ask 2 learners to answer using today's table.		question into data to collect. Note any learner still unsure about class boundaries for a short reteach at the start of Lesson 4.
Model Building Phase VIDEO: Grouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar videos EXERCISE: Grouped Data to Find the Mean Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Groups update their cumulative model (a poster with 'What we see in the data', 'What we still need to find out' and 'Our prediction for next steps'). They paste or copy the grouped frequency table and add: (1) Subject 1 clusters in 70-89 (15 of 20 learners) with no scores below 60; Subject 2 spreads from 10-19 to 60-69, with 16 of 20 below 50. (2) Revised prediction: Mr. Otieno's class may need extension work; Mrs. Wanjiku's class and the learners may need revision and support, but we need more data (for example topic-by-topic marks and a second test) to find the cause. (3) A sketch of how the grouped table could become a graph. Learners compare with their Predict Phase prediction and write one sentence on what changed. Groups present to another group in 1 minute.	Say: 'Use the grouped table as evidence in your model. What does it add to the ordered list?' WAIT TIME 10-15 seconds. Circulate and ask: 'Which class would you tell Madam Achieng to look at first? Why?' and 'Can we say the learners are weak?' Cold-call 3 groups to share one revision to their model. Insist that predictions are evidence-based and that causes are phrased as hypotheses ('may', 'could'). Give feedback on accuracy of frequencies and totals. Preview the next step: 'In Lesson 4 we will calculate the mean, median and mode from the ungrouped scores, so we can describe the centre of each subject with one number.' Keep the posters for Lesson 4.	Cumulative Model Revision: learners integrate new evidence (grouped tables showing clustering and spread) into their evolving model of how data supports decisions. Writing what changed from the original prediction builds metacognition and shows how evidence reshapes ideas.	Look for models that include correct frequencies for both subjects, mention clustering and spread, link to the pass mark, propose actions for both teachers and the learners, and state what further data are needed. Check that no group claims a cause as a fact. Indicators of mastery: learner can explain why the same classes are used for both subjects, and can state the class, boundaries and midpoint for any given score.

D. TEACHER REFLECTION

After the lesson, consider: Which learners confused class limits (10-19) with boundaries (9.5-19.5), and which needed to be shown where 20 or 60 belong? Did groups choose class width 10 for good reasons from the range, or just because it was suggested? Were tally counts accurate, and did every total reach 20? How many learners avoided stating causes as facts? Which questions on the Driving Question Board show readiness for the mean, median and mode in Lessons 4 and 5, and for graphs in Lessons 6 and 7? What would you change next time, for example more time for classifying or a shorter gallery walk?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	We organised the 20 scores of Subject 1 and Subject 2 into grouped frequency tables using a common class width of 10 (classes 10-19 to 90-99). Subject 1 frequencies: 0, 0, 0, 0, 0, 2, 7, 8, 3 and Subject 2 frequencies: 4, 4, 5, 3, 2, 2, 0, 0, 0, each totalling 20.
What did I learn?	We learned how to use the range (28 and 49) to choose a class width, write class limits, find class boundaries (9.5, 19.5, ..., 99.5) and midpoints (14.5, ..., 94.5), and classify scores with tally marks so that frequencies add to $n = 20$.
How does this explain the phenomenon?	Grouping shows that Subject 1 scores cluster high (70-89 has 15 of 20 learners and no score is below 60), while Subject 2 scores are spread out (16 of 20 are below 50). The raw lists hid this pattern. The tables do not tell us why, so we need more data, such as topic-by-topic marks and a second test, before deciding the causes and next steps.

LESSON 4 (80 minutes (2 x 40-minute periods)): Mean, Mode and Median of Ungrouped Data: Comparing the Two Subjects

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To help learners calculate and interpret the mean, mode and median of the ungrouped scores in Subject 1 and Subject 2, and use these single typical values to compare the two subjects and move closer to answering how the data can predict next steps.
Knowledge	Learners will be able to: (1) calculate the mean as sum of x divided by n (Subject 1: $1607/20 = 80.35$; Subject 2: $677/20 = 33.85$); (2) find the median as the average of the 10th and 11th ordered values (Subject 1: $(80+81)/2 = 80.5$; Subject 2: $(31+32)/2 = 31.5$); (3) find the mode from the ungrouped frequencies, including the no-mode case (Subject 1) and the bimodal case (Subject 2: 32 and 60); (4) interpret each measure in the context of the two subjects.
Skills	Calculating, ordering and counting data; analysing and interpreting data (NGSS-style practice: analysing and interpreting data); using mathematics and computational thinking; constructing explanations from evidence; communicating findings on manila paper; working collaboratively in groups of 4.
Attitudes	Curiosity about what numbers tell us; honesty and accuracy when calculating; respect for others' ideas during group discussion; caution in drawing conclusions (using words like 'may', 'could' and 'we need more data'); appreciation of data in making decisions.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	Lessons 1-3 showed that raw lists are hard to read: Lesson 1 ordered the scores and counted passes (20 pass in Subject 1; 4 pass and 16 fail in Subject 2), Lesson 2 built ungrouped frequency tables, and Lesson 3 grouped the data. Lesson 4 now summarises each whole set with ONE typical value (mean, median, mode). These exact values give the evidence that Lesson 5 will estimate again from grouped data, and that Lessons 6-7 will compare with graphs. This lesson gives learners their first numeric answer to: do the two subjects differ in their typical score, and what might Madam Achieng, Mr. Otieno and Mrs. Wanjiku do next?
Safety Notes	No hazardous materials. Learners use pencils, rulers, manila paper and calculators (or manual long addition). Keep walkways clear when groups display manila paper. Remind learners to treat the scores as class data and not to tease any individual learner about their marks (Learner 1 to Learner 20 are used instead of names).

B. LESSON OVERVIEW

Madam Achieng has asked the Grade 10 class at a day secondary school in Kisumu to use the scores of the same 20 learners in Subject 1 (Mr. Otieno) and Subject 2 (Mrs. Wanjiku) to decide next steps. In earlier lessons learners ordered the scores, counted passes and built frequency tables. In this lesson, groups of 4 predict which subject has the higher typical score, then calculate the mean, median and mode of each set from the ordered lists and ungrouped frequency tables. They discover that Subject 1 has mean 80.35, median 80.5 and no mode, while Subject 2 has mean 33.85, median 31.5 and two modes (32 and 60). They interpret what each measure says, compare the two subjects, and note what the measures cannot tell them (such as WHY the scores differ or how spread out they are). The Driving Question Board and the class model are updated with this new evidence.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners recall the puzzle: the same 20 learners scored 67 to 95 in Subject 1 but 11 to 60 in Subject 2. Each learner individually writes on a sticky note or in the exercise book: (a) 'If we had to choose ONE number to describe the whole class in Subject 1 and ONE for Subject 2, what would the numbers be, roughly?' (b) 'Will the one number for Subject 2 be above or below 50? Why?' Then in groups of 4 they share and agree on a group prediction, writing it on manila paper. They also predict: 'Will Subject 1 have a score that appears most often? Will Subject 2?' using the ascending lists from Lesson 1 displayed on the board.	Display the two ascending lists (Subject 1: 67, 69, 70, 72, 73, 74, 76, 77, 78, 80, 81, 82, 84, 85, 86, 88, 89, 90, 91, 95; Subject 2: 11, 13, 15, 18, 21, 23, 27, 29, 30, 31, 32, 32, 35, 42, 44, 45, 51, 58, 60, 60). Say: 'Madam Achieng cannot read 40 numbers every time she makes a decision. What single number could stand for each subject?' Give 2 minutes of silent individual writing. WAIT TIME: after asking 'Will the typical score in Subject 2 be above or below the pass mark of 50?' wait 10-15 seconds before taking responses. Cold-call 4 groups to share predictions; record ALL predictions on the board without judging. Ask: 'What in the lists made you choose that number?' Do not say whether predictions are right.	Prediction with reasoning (Predict-Observe-Explain): Learners commit to a prediction and justify it with evidence from the ordered lists. This surfaces the idea that 'typical' could mean the middle, the most common or the 'fair share' value, and creates a need for the three measures. Learners later check predictions against calculations.	Observe whether learners (1) refer to the lists when justifying predictions, (2) use words like 'middle', 'most common' or 'average' for 'typical', and (3) predict Subject 1 higher than Subject 2. Tally on a clipboard how many groups predict a Subject 2 typical value below 50. Note misconceptions such as 'the typical score is the highest score' or 'mean is always the middle value'.
Observe Phase  VIDEO: Statistics intro: Mean, median, & mode Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Station-style group investigation (groups of 4, each learner takes a role: reader, recorder, calculator, checker). Part A (MODE): Using the ungrouped frequency tables from Lesson 2, groups identify the score with the highest frequency. Subject 1: every score occurs once, so they must decide what that means (no mode). Subject 2: 32 and 60 each occur twice (bimodal). Part B (MEDIAN): Using the ascending lists, groups count to the 10th and 11th values, since $n = 20$ is even. Subject 1: 10th = 80, 11th = 81, median = $(80+81)/2 = 80.5$. Subject 2: 10th = 31, 11th = 32, median = 31.5. They cross-check by counting 10 values on each side. Part C (MEAN): Groups add all 20 scores in each subject (Subject 1 sum = 1607; Subject 2 sum = 677), then divide by 20:	Circulate with a checklist. Pose: 'Where in the table do you look to find the mode? What does a frequency of 1 for every score tell you?' For the median ask: 'Why do we need two middle values when $n = 20$? Show me with your pencil which are the 10th and 11th.' For the mean: 'Add in pairs or tens to reduce errors; check your sum with your partner. What should the sums be close to?' Provide a quick check: if a group's Subject 1 sum is not 1607 or Subject 2 sum is not 677, say 'Recheck the addition in two halves: first ten scores, then last ten'. WAIT TIME: after asking 'Can a data set have no mode? Two modes?' wait 10-15 seconds, then cold-call 3 learners from different groups. Remind them to keep the Learner 1 to Learner 20 order only for the raw lists, not for	Structured data collection and cross-checking (Analysing and Interpreting Data): Learners calculate the three measures using three different representations (frequency table, ordered list, sum), then verify each result by a second method. This builds understanding that the measures describe the SAME data in different ways and that checking guards against errors.	Observable indicators: (1) learners correctly identify the 10th and 11th values and average them; (2) learners state 'no mode' for Subject 1 rather than writing 0 or 80; (3) learners write both 32 and 60 as modes; (4) sums 1607 and 677 and means 80.35 and 33.85 appear on manila paper, with answers rounded to 2 dp. Use a thumbs-up/down check: 'Thumbs up if you got a sum of 1607 for Subject 1'. Note groups needing support with division by 20 and give a mini-conference.

	mean = 80.35 for Subject 1 and 33.85 for Subject 2. Groups record $\bar{x}$, sum of x and n in a results table on manila paper, and compare with their predictions.	the ordered lists. Use the Kenyan side example (mandazi sales in Mombasa: 20, 25, 25, 30, 40 mandazi, mean 28) briefly to show mean as 'fair share' if learners struggle.		
Explain Phase  VIDEO: Mean and standard deviation versus median and IQR <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating mean and median from data displays <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups use their results to build a comparison table (Measure Subject 1 Subject 2): mean 80.35 33.85; median 80.5 31.5; mode none 32 and 60. Then they write explanations answering: (1) 'What does the mean of 80.35 tell us about Subject 1 and the mean of 33.85 about Subject 2 compared with the pass mark of 50?' (2) 'Why are the mean and median close in Subject 1 and also close in Subject 2? What would happen to the mean if Learner 14, who scored 11 in Subject 2, had scored 51?' (3) 'Which measure is most useful for Madam Achieng? Why is the mode less useful here?' (4) 'What can these three numbers NOT tell us?' Learners should note that they cannot say WHY the scores differ (test difficulty, teaching method and revision time are unknown) and that the measures do not show spread (the range 28 versus 49 from Lesson 1 is still needed). Groups share explanations in a gallery walk, leaving one question on a sticky note for another group.	Say: 'Use evidence from your table: because the Subject 1 mean is 80.35, we can say...' Provide sentence starters: 'The typical score in Subject 1 is higher than in Subject 2 because...', 'This may mean... but we need more data about...'. Ask the 'what if' question about changing one score and give 15 seconds of WAIT TIME before cold-calling 3 learners; guide them to see that one extreme score moves the mean by $40/20 = 2$ marks while the median stays near the middle. Probe: 'Does a higher mean show that Mr. Otieno teaches better than Mrs. Wanjiku? What would we need to know?' Correct any causal claims by asking: 'Is that a fact from the data or a hypothesis to investigate?' Facilitate the gallery walk (5 minutes), asking each group to leave 1 'I wonder' sticky note. Do NOT introduce grouped-data formulas (Lesson 5), histograms (Lesson 6) or quartiles (Lessons 7-8).	Claim-Evidence-Reasoning with a Gallery Walk: Learners make a claim about the subjects, support it with the calculated measures, and reason about what the measures show and hide. Peer review during the gallery walk exposes different interpretations and builds the habit of caution about causes (hypotheses, not facts).	Look for explanations that (1) cite specific numbers (80.35, 33.85, 80.5, 31.5) rather than 'higher' only; (2) compare to the pass mark of 50; (3) use cautious language ('may', 'could', 'we need more data'); (4) recognise that mean and median can both describe the centre but may differ with extreme scores. Use an exit-style question on a half sheet: 'Write one sentence comparing the typical score in the two subjects and one thing the typical score cannot tell us.' Sort responses into: accurate, partly accurate, needs reteaching.
Driving Question Board (DQB) Creation  VIDEO: Ungrouped Data to Find the Mean Principles <i>Related topic: not an exact match for this lesson.</i>	Learners revisit the Driving Question Board with the question: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' In groups they (a) place sticky notes showing what they now KNOW (for example, 'Typical score in Subject 1 is about 80 and in Subject 2 about 32-34', 'Mean and median are close in each subject', 'Subject	Read the Driving Question aloud, then say: 'Look at the board. What have we found out today that helps with next steps? What new questions do we have?' Give 1 minute of silent writing and WAIT TIME of 10-15 seconds before asking for volunteers. Cold-call 3 groups to place and justify notes. Sort questions into categories on the board: 'Spread', 'Graphs',	Driving Question Board as a public record of learning: Learners track what they know, what they still wonder, and where they will look next. This makes progress visible and keeps the sequence of lessons purposeful, linking each calculation back to the decision Madam Achieng must make.	Check that sticky notes include at least one evidence-based statement with a number from the lesson and at least one new question about spread, causes or action. Observe whether learners place notes in the right category. Count the groups able to link a question to a future lesson. Record learners who only write vague questions ('Why?') and prompt with

Source: CK-12 Search ARES for similar videos HTML: Stem-and-Leaf Plots, Mean, Median and Mode (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	1 has no mode and Subject 2 has two modes'), (b) add new questions (for example, 'Why are the Subject 2 scores so much lower for the same learners?', 'How spread out are the scores around the mean?', 'Can we show this on a graph?', 'Do the 16 learners below 50 need remedial support?'), and (c) move earlier questions that are now partly answered. They link questions to the two KICD inquiry questions.	'Causes (need more data)', 'Actions'. Mark questions Lesson 5 (grouped estimates), Lesson 6-7 (graphs) or Lesson 8 (interpretation) will address. Highlight that the question about causes cannot be answered from these scores alone and may need topic-by-topic marks or a second test, as Mr. Otieno and Mrs. Wanjiku might plan.		'Why what? Which number made you wonder?'
Model Building Phase VIDEO: Impact on median & mean: increasing an outlier <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Stem and Leaf Plots, Mean, Median, and Mode Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners update the class model (a large poster showing the pathway from raw data to decision). In groups they add a new layer labelled 'Summarise with one typical value' showing: raw scores -> ordered list -> frequency table -> grouped table -> mean, median, mode -> (coming next: graphs and spread) -> comparison -> next steps. Beside it they write the key findings: Subject 1: mean 80.35, median 80.5, no mode; Subject 2: mean 33.85, median 31.5, modes 32 and 60. They add a first tentative prediction for next steps, labelled as a hypothesis: for example, 'Subject 1 learners are passing at a typical level well above 50; Subject 2 typical score is below 50, so revision may be needed, but we need more data to know why.' Each learner completes a personal 'What I observed / What I learned / How I can explain it' entry for the Summary Table.	Say: 'Show on the model where today's work fits. What new arrow or box do we need?' Ask: 'What does the model still NOT explain?' (spread, shape, causes). Give WAIT TIME of 10-15 seconds, then cold-call 3 learners to read their entries. Guide learners to word predictions cautiously: 'Mr. Otieno's class may continue with extension work; Mrs. Wanjiku and the class may need to revise Subject 2 topics; we need more data to find the cause.' Do not give final recommendations; those are developed in Lessons 8-9. Collect Summary Table entries for review. Set a short take-home task: find the mean of five daily mandazi sales or tea-picking weights from a local context to practise the method.	Cumulative model revision: Learners add the new evidence to a shared model and identify gaps, so each lesson builds on the last. Writing tentative predictions as hypotheses develops scientific habits of caution and evidence-based reasoning.	Check that the model includes the correct values (80.35, 80.5, none; 33.85, 31.5, 32 and 60) and that the predictions use cautious language. Observable indicators: learners can explain in their own words what mean, median and mode each tell us, and name one limitation (no information about spread or causes). Review Summary Table entries for accuracy, and note learners to support in the next lesson on grouped data.

D. TEACHER REFLECTION

After the lesson reflect: (1) Did every group correctly use the 10th and 11th values for the median and write 'no mode' and 'bimodal' correctly? Which learners confused

no mode with a mode of zero? (2) Did learners use cautious language when interpreting the difference between subjects, or did they claim causes such as 'Mr. Otieno teaches better'? How did I respond? (3) How well did learners connect the measures to the pass mark of 50 and to Madam Achieng's decision? (4) Which addition or division errors were most common, and should I provide a quick calculation routine before Lesson 5? (5) Which DQB questions about spread and graphs will drive the next lessons, and did learners notice that mean and median alone do not show how spread out the scores are? (6) Were wait time and cold-calling used evenly so quieter learners contributed?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Calculating from the ordered lists and frequency tables, we observed that Subject 1 has a sum of 1607, mean 80.35, median 80.5 and no mode, while Subject 2 has a sum of 677, mean 33.85, median 31.5 and two modes (32 and 60).
What did I learn?	We learned how to calculate the mean (sum of x divided by n), the median (average of the 10th and 11th ordered values when n = 20) and the mode (the most frequent score, which may be none or more than one), and that each gives a different way to describe a typical score.
How does this explain the phenomenon?	The typical score in Subject 1 (about 80) is well above the pass mark of 50, while the typical score in Subject 2 (about 32-34) is below it, so the difference seems to be between subjects and not because the learners are weak overall; the measures may help predict next steps but they do not show spread or the cause, so we need more data.

LESSON 5 (80 minutes (2 x 40-minute lessons)): Mean, Mode and Median of Grouped Data: Estimating the Centre of Each Subject

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners estimate the mean, mode and median of Subject 1 and Subject 2 from the grouped frequency tables, compare the estimates with the exact ungrouped values from Lesson 4, and explain why grouped measures are only estimates. This moves the class closer to Madam Achieng's request to predict next steps for the learners, Mr. Otieno and Mrs. Wanjiku.
Knowledge	Learners will know that: (1) the estimated mean of grouped data = $\text{sum of } fx / \text{sum of } f$, using class midpoints x ; (2) the modal class is the class with the highest frequency, and the mode is estimated by $L + (d_1/(d_1+d_2)) \times w$; (3) the median class contains the $(n/2)$ th value, and the median is estimated by $L + ((n/2 - cf)/f) \times w$; (4) grouped measures are estimates because individual scores are replaced by midpoints or positions within classes, so they differ slightly from exact values (Subject 1 exact mean 80.35, median 80.5; Subject 2 exact mean 33.85, median 31.5).
Skills	Learners will: calculate midpoints and fx ; sum fx and divide by $n = 20$; identify modal and median classes; apply the mode and median formulas with class boundaries; compare estimates with the exact values; communicate results in a comparison table. Science and Engineering Practices (adapted for Mathematics): Analysing and Interpreting Data; Using Mathematics and Computational Thinking; Constructing Explanations; Engaging in Argument from Evidence.
Attitudes	Learners will show accuracy and patience when working with several steps, honesty in reporting estimates as estimates, curiosity about why two subjects differ, respect for others' methods in groups of 4, and caution in not claiming reasons for the score difference without more data.
Key Inquiry Question	How do we organise data using frequency distribution tables? Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	Lesson 3 gave us the grouped tables with boundaries and midpoints. Lesson 4 gave us the exact mean, median and mode from the ungrouped lists. In Lesson 5 learners use only the grouped tables to estimate the same measures, and learn how close the estimates are. These estimates give the centre of each subject (about 80 versus about 34) and prepare for Lesson 6 (histograms and frequency polygons, using the modal class and midpoints) and Lesson 7 (ogives, with the grouped median formula as a check).
Safety Notes	Low-risk classroom lesson. Learners work on desks and manila paper; keep walkways clear. Remind learners that the marks belong to Learner 1 to Learner 20 and that scores should not be used to tease anyone. Groups of 4 should share tasks fairly so that every member calculates.

B. LESSON OVERVIEW

Madam Achieng, the Deputy Principal of a day secondary school in Kisumu, asked the class to use the scores of the same 20 learners in Subject 1 (Mr. Otieno) and Subject 2 (Mrs. Wanjiku) to decide next steps. In Lessons 1-4 learners ordered the scores, built tables and found exact measures. Today they work only from the grouped tables (classes 10-19 to 90-99) to estimate the mean, modal class and mode, and median. They predict how close grouped estimates will be to the exact values, calculate them step by step, and compare. The estimates (means 80.5 and 35; modes 81.2 and 32.8; medians 80.75 and 33.5) agree closely with the exact values, so the picture is clear: Subject 1 is high, and Subject 2 is low. Learners also see why the numbers are estimates. The lesson ends by adding the estimates to the Driving Question Board and to the class model. We still do not know WHY the scores differ; reasons remain hypotheses to investigate.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Grouped Data to Find the Mean Principles Source: CK-12  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the grouped frequency table from Lesson 3 (classes 10-19 to 90-99 for both subjects) and the exact measures from Lesson 4 (Subject 1 mean 80.35, median 80.5, no mode; Subject 2 mean 33.85, median 31.5, modes 32 and 60). Individually they write: 'If Madam Achieng only gave us the grouped table and not the 20 scores, could we still find the mean, mode and median? Would they be exactly the same? Predict the grouped mean for each subject.' They record a prediction on a sticky note or in their books (e.g. 'about 80 for Subject 1 and about 35 for Subject 2') with a reason, then share with a partner.	Display the grouped table on the board (Subject 1 frequencies 0,0,0,0,0,2,7,8,3; Subject 2 frequencies 4,4,5,3,2,2,0,0,0). Say: 'Imagine the 20 scores are lost and only this table survives. Can we still say where the centre of each subject is?' Give 1 minute silent writing, then WAIT TIME of 10-15 seconds before any talk. Pair-share for 2 minutes. Cold-call 4 learners: 'What do you predict for Subject 1? For Subject 2? Why?' Ask 2 more: 'Will it match 80.35 and 33.85 exactly?' Record predictions on the board without judging them. Say: 'We will test your predictions with evidence today.'	Predict-Observe-Explain with a Prediction Gallery: committing to a numerical prediction before calculation creates a reason to calculate and exposes the idea that a table hides individual scores. Learners compare their predictions with results later, which builds understanding of why estimates occur.	Observe: (1) Do predictions place Subject 1 near the 70s-80s and Subject 2 near the 30s-40s using the table? (2) Do learners give a reason (e.g. 'most scores are in 80-89') rather than a guess? (3) Do at least 5 learners say the answers may not be exactly equal to Lesson 4's values? Note learners who think grouping loses no information and revisit this in the Explain Phase.
Observe Phase  VIDEO: Grouped Data to Find the Mean Principles Source: CK-12  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice Source: CK-12  Search ARES for similar readings	Part A: Estimated mean. In groups of 4 learners copy the grouped table and add columns x (midpoint: 14.5, 24.5, 34.5, 44.5, 54.5, 64.5, 74.5, 84.5, 94.5) and fx . For Subject 1 they calculate $2(64.5)+7(74.5)+8(84.5)+3(94.5) = 129+521.5+676+283.5 = 1610$ and mean $= 1610/20 = 80.5$. For Subject 2 they calculate $4(14.5)+4(24.5)+5(34.5)+3(44.5)+2(54.5)+2(64.5) = 58+98+172.5+133.5+109+129 = 700$ and mean $= 700/20 = 35$. Part B: Modal class and mode. They find the modal classes (Subject 1: 80-89, $f = 8$; Subject 2: 30-39, $f = 5$), then use boundaries: Subject 1: $L = 79.5$, $d1 = 8-7 = 1$, $d2 = 8-3 = 5$, mode $= 79.5 + (1/6)(10) = 81.2$; Subject 2: $L = 29.5$, $d1 = 5-4 = 1$, $d2 = 5-3 = 2$, mode $= 29.5 + (1/3)(10) = 32.8$. Part C: Median. They	Circulate with a checklist. Launch with: 'Why do we use the midpoint of 60-69 as 64.5 and not 67 or 69?' WAIT TIME 10-15 seconds, then cold-call 2 groups. Model ONE row (Subject 1, class 70-79: $7 \times 74.5 = 521.5$) on the board, then let groups continue. Prompt: 'What does the sum of f have to be? Check it equals 20 before you go on.' For the mode, say: 'Which class is tallest? What are the classes before and after it? Write $d1$ and $d2$ BEFORE you substitute.' For the median say: 'The median is the 10th value. How many learners are below 79.5? Then how far into the 80-89 class do we need to go?' Cold-call 3 learners to give the median class and cf . Stop each group at the end of each part to check answers (1610, 700; 81.2, 32.8; 80.75,	Structured Data Analysis with a Worked Example and Think-Aloud: one modelled row followed by learner calculations lets learners see the procedure and then own it. Comparing grouped estimates with exact Lesson 4 values builds a sense of accuracy and develops the concept of 'estimate'.	Check: (1) midpoints written correctly for all 9 classes; (2) sum of $fx = 1610$ and 700 and divided by 20 (not by 9, the number of classes); (3) correct boundaries used (79.5 and 29.5), not limits (80 and 30); (4) $d1$ and $d2$ identified correctly; (5) correct cf before the median class (9 and 8). Common errors to note: dividing by the number of classes; using the class limit instead of the boundary; using cf of the median class instead of the cf before it; confusing the modal class with the mode. Collect one group's table as a quick check.

	build a cumulative frequency column (Subject 1: 0,0,0,0,0,2,9,17,20; Subject 2: 4,8,13,16,18,20,20,20,20), find the 10th position ($n/2 = 10$), the median class (Subject 1: 80-89 with cf before = 9, $f = 8$; Subject 2: 30-39 with cf before = 8, $f = 5$), and calculate: Subject 1 = $79.5 + (1/8)(10) = 80.75$; Subject 2 = $29.5 + (2/5)(10) = 33.5$. Learners complete a comparison table: grouped estimate versus exact value from Lesson 4.	33.5) before moving on. Point out that zero-frequency classes contribute $fx = 0$ but are still in the table. Extension question: 'If the class width were 5, would the estimates change?'		
Explain Phase  VIDEO: Statistics intro: Mean, median, & mode <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Grouped Data to Find the Mean (Flexbook) Source: CK-12  Search ARES for similar readings	Groups complete a comparison table: Measure Subject 1 exact (Lesson 4) Subject 1 grouped estimate Subject 2 exact Subject 2 grouped estimate. Mean: 80.35 / 80.5; 33.85 / 35. Median: 80.5 / 80.75; 31.5 / 33.5. Mode: none / 81.2 (modal class 80-89); 32 and 60 / 32.8 (modal class 30-39). They then discuss: (a) Which estimates are close to the exact values? (b) Why are they not exactly equal? (e.g. grouping replaces each score with the midpoint of its class; the actual scores in 80-89 range from 80 to 89 and average 84.1, not 84.5; the formulas assume scores are evenly spread within the class). (c) Subject 1 has no exact mode, yet the grouped table gives a modal class and mode estimate. What does this tell us? (d) What do the estimates say about the two subjects and what could Madam Achieng predict? Each group writes a 2-3 sentence claim-evidence-reasoning statement on manila paper, e.g. 'Subject 1 estimated mean 80.5 is much higher than Subject 2 estimated mean 35, so the typical learner scores far higher in Subject 1. We	Pose: 'Is 80.5 a better or worse answer than 80.35? Which one is exact?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Press for reasoning: 'Say that again using the words midpoint and class.' Guide a mini-demonstration: Subject 1 class 60-69 holds scores 67 and 69, but we treated them as 64.5 and 64.5. 'Who is that unfair to? Who is helped?' Then ask: 'Would it change the decision if the mean were 80.35 or 80.5?' (No: both show all learners well above 50.) Challenge a misconception: 'Someone said the grouped mean is wrong. Is it wrong or an estimate?' Ask 4 cold-call learners to read out their claim statements. Insist on cautious language: 'Can we say Mrs. Wanjiku taught badly? What do we actually know? What do we NOT know?' Record: 'Estimates close to exact values; difference between subjects is clear; reasons unknown, so we need more data.'	Claim-Evidence-Reasoning (CER) with Error Analysis: learners state a claim (the estimates compare well and show a big gap), cite the numbers as evidence, and reason about how grouping causes small differences. This builds a deeper understanding of estimation and develops statistical caution about causes.	Look for: (1) learners say estimates differ because midpoints and position formulas stand in for actual scores; (2) learners do not say the estimates are mistakes; (3) CER statements cite at least two numbers (e.g. 80.5 and 35); (4) statements avoid claiming causes and use 'may', 'could' or 'we need more data'; (5) learners state that the grouped values lead to the same conclusion as the exact values. Use a thumbs-up/sideways/down check on: 'I can explain why grouped measures are estimates.'

	need more data to find out why.'			
Driving Question Board (DQB) Creation  VIDEO: Estimating mean and median in data displays <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Grouped Data to Find the Mean Practice Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' On sticky notes they add (a) one thing they now know, e.g. 'The estimated means are 80.5 and 35, so Subject 1 is much higher' and (b) one new question, e.g. 'Can a graph show the spread better than these measures?', 'Both subjects are centred differently, but is one more spread out?', 'Why did the same learners do so well in one subject and so poorly in the other?', 'Does the mean hide the 16 learners below 50?' They place notes under headings: Answered / Partly answered / New questions. Groups link at least one note to the Key Inquiry Questions on organising data with tables and on why collecting, organising and interpreting data matters for decisions.	Say: 'Look at the board. What has today's evidence added to our answer for Madam Achieng?' WAIT TIME 10-15 seconds. Cold-call 3 learners to read a new note aloud and say which earlier note it connects to. Cluster questions on the board: 'centre', 'spread', 'causes', 'graphs', 'action'. Point out: 'Questions about spread and shape need graphs, which we will draw in Lesson 6 and 7. Questions about why need new data, which we will plan in Lesson 9.' Move the question 'Are the grouped values the same as the exact ones?' to 'Answered: close, but estimates'. Keep any claim about causes under 'Questions to investigate'.	Driving Question Board with Question Clustering: externalising what is known and not yet known keeps the investigation learner-driven, connects today's calculation to the anchoring phenomenon, and shows how the next lessons (graphs and quartiles) respond to learners' own questions.	Check: (1) at least one note per learner cites a specific number from today (80.5, 35, 81.2, 32.8, 80.75, 33.5); (2) questions are genuine and linked to the phenomenon (spread, graphs, causes, action); (3) at least 3 groups raise a question about spread or the 16 learners below 50; (4) no note states a cause as a fact. Record themes to adjust Lesson 6.
Model Building Phase  VIDEO: Free response example: Significance test for a mean <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode	Learners revise the class model for the driving question, using a summary table on manila paper with columns: Subject Estimated mean Modal class / estimated mode Median estimate Prediction (cautious). Subject 1: 80.5 80-89 / 81.2 80.75 'Scores are high around the 80s; all learners are likely to pass again; extension work could be given.' Subject 2: 35 30-39 / 32.8 33.5 'Typical score is in the 30s, below the pass mark of 50; revision and remedial support are likely needed; we need more data (topic-by-topic marks, a second test) to find causes.' Learners add a line: 'Grouped measures are estimates	Say: 'Let us add today's evidence to our model for Madam Achieng. What is the typical score in each subject?' WAIT TIME 10-15 seconds. Cold-call 4 learners to complete the table cells orally before groups write. Ask: 'Which measure would you give to Madam Achieng first, and why?' Prompt careful predictions: 'Which words show we are not sure of the cause: may, could, or must?' Ask 2 groups to present their table on manila paper; require them to name one strength and one limitation of grouped estimates. Use the exit ticket to sort learners into 'secure' (all steps correct), 'nearly' (one slip), and 'not yet',	Cumulative Model Revision with Gallery Walk: updating the shared model with new evidence from the grouped measures shows how understanding builds lesson by lesson. Writing a cautious prediction trains learners to separate what the data show (centre, gap) from what they do not show (causes).	Success looks like: (1) tables carry correct values (80.5, 35; 80-89 and 30-39; 81.2 and 32.8; 80.75 and 33.5); (2) predictions are tied to numbers and the pass mark of 50; (3) learners use 'estimate' for grouped values; (4) hypotheses are separate from facts; (5) the exit ticket shows correct midpoints, fx and division by n. Target at least 15 of 20 learners secure on the estimated mean; follow up with a small group on the median formula if more than 5 learners misuse cf.

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar readings	and differ slightly from the exact values (80.35, 33.85; 80.5, 31.5).' Each group adds one 'What we still need' arrow: 'spread picture (graphs, Lesson 6 and 7)'. Learners then complete a quick exit ticket: 'Calculate the estimated mean for a small table and say why it is an estimate.' Optional side example: daily mandazi sales in Mombasa grouped in classes of 10 (this is a practice context only and does not change the anchor data).	and plan a short revisit for the last group. Preview: 'Next lesson we draw histograms and frequency polygons to SEE these two subjects side by side, using the same classes and scales.'		
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D. TEACHER REFLECTION

Reflect after the lesson: (1) Did learners divide sum of fx by $n = 20$ rather than by 9 classes? (2) How many used boundaries (79.5, 29.5) correctly in the mode and median formulas? (3) Could learners explain why 80.5 and 80.35 differ without saying one is wrong? (4) Did any learner or group claim a cause for the difference between Subject 1 and Subject 2? How did I respond? (5) Which questions on the DQB point to spread and graphs, and how will I use them to launch Lesson 6? (6) Which learners need extra practice with cumulative frequency before the ogive work in Lesson 7? Consider whether pacing across the two 40-minute periods allowed enough time for Part C (median).

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the grouped tables let us estimate the mean (Subject 1: $1610/20 = 80.5$; Subject 2: $700/20 = 35$), the modal class and mode (80-89, about 81.2; 30-39, about 32.8) and the median (about 80.75 and 33.5). These are close to the exact values from Lesson 4 but not identical.
What did I learn?	We learned how to use midpoints and fx to estimate the mean, boundaries and $d1$, $d2$ to estimate the mode, and cumulative frequency to estimate the median. We learned that grouped measures are estimates because each score is replaced by its class midpoint or position in the class.
How does this explain the phenomenon?	We can explain that Subject 1 is centred around 80 and Subject 2 around 34, so decisions must be made per subject: the same learners cannot all be 'weak'. We cannot yet say why the scores differ, and we need spread information and more data before deciding the full next steps.

LESSON 6 (80 minutes (2 x 40-minute lessons)): Representation of Data I: Histograms and Frequency Polygons

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To draw a histogram using class boundaries and a frequency polygon using class midpoints for Subject 1 and Subject 2, and to read the shape and spread of each subject from the graphs.
Knowledge	Learners draw a histogram with touching bars at the class boundaries (9.5 to 99.5) with frequency on the vertical axis. They draw a frequency polygon by plotting (midpoint, frequency) points at 14.5, 24.5, ..., 94.5 and joining them with straight lines, closing to the axis at the empty classes either side. They know both graphs must use the same classes and the same axis scales so that the two subjects can be compared.
Skills	Drawing and labelling graphs accurately on graph paper; reading the grouped frequency table; choosing a scale; describing shape (clustered or spread out), modal class and spread; comparing two graphs; communicating findings on manila paper (groups of 4). Science and Engineering Practices: analysing and interpreting data, using mathematics, constructing explanations.
Attitudes	Accuracy and neatness in drawing; curiosity about what a picture of data shows; respect for others' ideas during groupwork; honesty in not claiming causes the data cannot show.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	Lessons 1 to 5 ordered the scores, built ungrouped and grouped tables, and calculated mean, mode and median. Lesson 6 turns the grouped table into pictures so that the shape and spread that the raw lists hid become visible. It prepares for Lesson 7 (cumulative frequency and ogives), which will reuse the graph-drawing skills, and Lesson 8 (comparing subjects and predicting next steps).
Safety Notes	Learners use sharp pencils, rulers and compasses/dividers carefully. Keep bags and walkways clear in the classroom. Groups pass manila paper and markers carefully and keep the floor free to prevent trips.

B. LESSON OVERVIEW

Back at the Grade 10 class in Kisumu, Madam Achieng still wants to know what to do about Mr. Otieno's Subject 1 and Mrs. Wanjiku's Subject 2. Learners have a grouped frequency table (common class width 10, classes 10-19 to 90-99, $n = 20$ for both subjects) and know the grouped means 80.5 and 35 and the modal classes 80-89 and 30-39. In this lesson they turn the table into a histogram and a frequency polygon for each subject on the same axes. They see that Subject 1 is a tight block of bars between 60 and 99, while Subject 2 is spread across 10 to 69 and sits much further left. Learners make claims from the pictures, but treat reasons for the difference as hypotheses to investigate and not as facts, because the data cannot tell us why.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Creating a	Learners look at the grouped frequency table for both subjects (classes 10-19 to 90-99; Subject 1:	Display the table on the board. Say: 'Madam Achieng cannot read 40 numbers quickly. Before we	Predict-Observe-Explain with Think-Pair-Share: committing to a prediction from the table first	Look for: sketches that place the Subject 1 tall bars on the right and the Subject 2 bars on the left

histogram Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Represent Real-World Data Using Bar Graphs, Frequency Tables and Histograms (Flexbook) Source: CK-12 Search ARES for similar readings	0, 0, 0, 0, 0, 2, 7, 8, 3; Subject 2: 4, 4, 5, 3, 2, 2, 0, 0, 0). Individually, for 3 minutes, they sketch what they think a bar graph of each subject would look like (where the tall bars will be, how wide the picture is) and write one sentence: 'I think the Subject 1 picture will look ____ and the Subject 2 picture will look ____ because ____.' They then share with a partner and the pair decides which subject will look more spread out.	draw anything, sketch what you expect each subject to look like.' Give 3 minutes of silent sketching. Then ask: 'Which subject will have its tallest bar on the right side of the graph, and why?' WAIT TIME 10-15 seconds before taking answers. Cold-call 4 learners, including at least 2 who did not volunteer. Ask each: 'What in the table makes you say that?' Record 2 or 3 predictions on the board without judging them. Ask: 'How will we know if your sketch is right?'	exposes learners' ideas about shape and spread. Later comparison of the drawn graphs with their sketches creates cognitive conflict or confirmation, which builds understanding of how a graph summarises a frequency table.	(correct use of the table); learners who refer to frequencies such as 8 in 80-89 or 5 in 30-39; learners who say a graph will show 'spread' without defining it. Note learners who sketch the bars with gaps or draw 20 separate bars, which shows they have not linked the table classes to a graph. Tick on a class list those who give a justified prediction.
Observe Phase VIDEO: Frequency Polygons Principles Source: CK-12 Search ARES for similar videos HTML: Constructing a Frequency Polygon (PLIX) Source: CK-12 Search ARES for similar readings	In groups of 4, learners draw on graph paper: (1) a histogram for Subject 1 and a histogram for Subject 2, using the same horizontal scale (class boundaries 9.5, 19.5, ..., 99.5) and the same vertical scale (frequency 0 to 8). The bars touch, are drawn at the class boundaries, and the classes with frequency 0 have no height. (2) A frequency polygon for each subject on the same axes: they find the midpoints 14.5, 24.5, 34.5, 44.5, 54.5, 64.5, 74.5, 84.5, 94.5, plot (midpoint, frequency) points, join with straight lines and close the polygon to the axis at the empty classes on either side (4.5 and 104.5). Roles: scribe, plotter, checker (checks the boundaries and frequency against the table), reporter. Learners check that bar heights add to 20 for each subject. Each group draws both polygons on one pair of axes in two colours to compare.	Model the first bar only: 'The class 10-19 really starts at 9.5 and ends at 19.5 because scores are whole numbers and the bars must touch. Where does the bar for 20-29 start?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Circulate and ask: 'Why is there no bar for 50-59 in Subject 1?' and 'What is the midpoint of 60-69 and why do we plot it there?' Prompt: 'What is the sum of your heights? It must be 20.' For groups that finish early: 'Where do you put the first and last point of the polygon, and why do we close it to the axis?' Stop at 25 minutes for a gallery walk of 5 minutes where groups compare their graphs to the manila paper of another group and fix errors. Remind: 'Same scales for both subjects, or the comparison is not fair.'	Gallery Walk with Peer Checking: learners compare their drawn graphs with those of other groups, using a checklist (touching bars, boundaries, midpoints, equal scales, title and labels). This builds understanding through noticing and correcting the connections between the grouped table, class boundaries and midpoints from Lessons 3 and 5.	Observable indicators: bars drawn from 9.5 to 19.5 and so on (not 10 to 19) with no gaps; frequency (not score) on the vertical axis; heights of Subject 1 bars 2, 7, 8, 3 from 59.5 to 99.5 and Subject 2 bars 4, 4, 5, 3, 2, 2 from 9.5 to 69.5; polygon points at the midpoints and closed at the axis; the same scale on both graphs. Use a checklist to mark each group. Common errors to note: gaps between bars, plotting the polygon at class limits (like 10 or 19), wrong closing, and using different scales.
Explain Phase VIDEO: Histograms	Each group uses its two graphs to write on manila paper a short report for Madam Achieng,	Ask each group to present for 1 minute. Prompt with: 'Point to the evidence on your graph for what	Claim-Evidence-Reasoning with Whole-Class Discussion: groups state a claim, point to evidence on	Listen for: correct modal classes 80-89 and 30-39; use of the words 'spread' or 'cluster' with reference

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Frequency Polygons (Flexbook) Source: CK-12 Search ARES for similar readings	answering: (a) Where is the tallest bar (modal class) in each subject? (b) Which subject is more spread out and how do you know from the graph? (c) Which polygon is further to the right? What does this suggest about the scores? (d) What can the graphs NOT tell us (for example why the scores differ)? Learners compare these findings with their Predict sketches and explain any differences. Expected findings: Subject 1 modal class 80-89 ($f = 8$), scores clustered between 60 and 99, range 28; Subject 2 modal class 30-39 ($f = 5$), scores spread from 11 to 60 (range 49), only 4 of 20 above the pass mark of 50 (the classes 50-59 and 60-69 hold 4 learners). Two polygons hardly overlap (only at the class 60-69). The graphs show the difference is between subjects, as the same 20 learners are in both.	you are claiming.' After each report ask: 'Who can add to this, or disagree?' WAIT TIME 10-15 seconds. Cold-call 5 learners overall to restate another group's claim in their own words. Challenge: 'A learner says: Mrs. Wanjiku taught badly because Subject 2 is low. Does the graph prove this?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Guide: 'The graph shows what happened, not why. The test difficulty, teaching method and revision time are unknown. We call these hypotheses to investigate.' Ask: 'Which is easier to compare the two subjects, the histogram or the polygon, and why?' (the polygon puts both on one pair of axes). Link back: 'How did the graphs show patterns that the raw lists of 20 scores hid?'	the graph and give reasoning. Separating what the data show from what we only suspect builds scientific reasoning and honest interpretation of data.	to the graph; a claim that compares both subjects; statements like 'the graph cannot tell us why'. Red flags: claiming the cause of the low scores; saying the learners are 'weak' in general (the same learners are in both subjects); confusing frequency with score. Use a 3-point rubric (claim, evidence from the graph, appropriate caution) on each group's manila paper.
Driving Question Board (DQB) Creation VIDEO: Creating a histogram Source: Khan Academy (English - US curriculum) Search ARES for similar videos EXERCISE: Frequency Polygons Practice Source: CK-12 Search ARES for similar readings	Learners revisit the Driving Question Board: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' They add to the board (on sticky notes) (1) what the graphs helped answer, such as 'Subject 2 needs help for the learners below 50' and (2) new questions, for example 'How can we tell the pass rate from a graph?', 'What would a second test look like on the same graph?', 'How do we find the median and quartiles from a graph?' Learners sort the sticky notes into 'Answered by Lesson 6', 'Still to find out' and 'Need more data'.	Say: 'Look at the board. Which of our questions can the histograms and polygons help to answer now?' WAIT TIME 10-15 seconds. Cold-call 4 learners to place a note and explain why. Ask: 'What new question has the picture made you ask?' Take 3 volunteers and 2 cold-calls. Point out that the question about medians and quartiles from a graph leads to the next lesson (cumulative frequency and ogives). Ask: 'What data would we need to find out why Subject 2 is lower?' to reinforce that causes need more data (for example topic-by-topic marks, a second test).	Driving Question Board with Question Sorting: sorting learner questions links new evidence to the driving question, shows what is answered and what is not, and gives learners ownership of the next steps of the investigation.	Check that at least 8 notes show a link between the graphs and the driving question; look for questions about reading shape or spread, cumulative totals or further data. Note learners who still ask only 'why did they score low?' without suggesting data to collect. Record the sorted categories on the board for Lesson 7.
Model Building	Learners update the class model for the driving question in their	Say: 'Add today's evidence to our model. What does the picture	Cumulative Model Revision: adding graphs to the model lets	Exit-ticket indicator: learners state that the same axes and scales

Phase  VIDEO: Frequency Polygons Principles Source: CK-12  Search ARES for similar videos  HTML: Frequency Tables and Histograms (Flexbook) Source: CK-12  Search ARES for similar readings	exercise books or on a large chart. The model so far includes: ordered lists and pass/fail counts, grouped table, mean/median/mode. They add a new section 'Pictures of the data': a sketch of both histograms or the two polygons on the same axes, with labels 'Subject 1: high, tightly clustered, modal class 80-89' and 'Subject 2: low, widely spread, modal class 30-39'. They write 2 sentences: 'The graphs show that _____. This suggests the next step is _____, but we need more data to find _____.' Exit ticket: 'Why must the two polygons use the same axes scales?'	show that the table alone did not?' WAIT TIME 10-15 seconds. Cold-call 3 learners to read their sentences. Prompt: 'Which next step for Mr. Otieno's class and which for Mrs. Wanjiku's class does the graph support?' Expected: Subject 1 learners continue with extension or enrichment work; for Subject 2, reteach and run remedial support for learners below 50. Add: 'Why do we still need a second test or topic marks?' Collect exit tickets and check them. Assign homework: draw the frequency polygon for the daily mandazi sales in Mombasa (a short grouped table given on the board) to practise boundaries and midpoints; the data must not change the anchor data.	learners link earlier evidence (tables, measures) with a visual representation, strengthening their understanding of how organised data supports decisions.	make the comparison fair, or that unequal scales would give a misleading picture. Check the model for the correct labels (modal classes 80-89 and 30-39), a sentence about spread, and an appropriate statement about needing more data. Record learners needing support to redraw boundaries or midpoints for a short intervention before Lesson 7.
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D. TEACHER REFLECTION

Which learners could draw bars at the boundaries (9.5 to 99.5) and plot the polygon at midpoints (14.5 to 94.5) without help, and which still used class limits or left gaps? Did learners distinguish what the graphs show (shape, spread, modal class) from why the scores differ? How did the same-axes comparison change learners' predictions? What support do I need to give before Lesson 7 so that learners can use their graph skills to build the cumulative frequency table and ogive?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We drew histograms (bars touching at the boundaries 9.5 to 99.5) and frequency polygons (points at midpoints 14.5 to 94.5, closed to the axis) for Subject 1 and Subject 2 on the same axes. Subject 1: scores clustered between 60 and 99, tallest bar 80-89 ($f = 8$). Subject 2: scores spread from 10 to 69, tallest bar 30-39 ($f = 5$).
What did I learn?	A histogram shows the frequency of each class with bars at the class boundaries; a frequency polygon joins (midpoint, frequency) points and is closed at the empty classes either side. Using the same classes and scales for both subjects makes the comparison fair, and the graphs make shape and spread easy to see.
How does this explain the phenomenon?	The graphs show that Subject 1 is high and tightly clustered while Subject 2 is low and widely spread, so the difference is between subjects and not about the learners all being weak. The graphs cannot tell us why the scores differ, so we need more data (for example topic-by-topic marks and a second test) before concluding.

LESSON 7 (80 minutes (2 x 40-minute periods)): Representation of Data II: Cumulative Frequency and Ogives

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To build a cumulative frequency table for each subject, draw an ogive, and read off the median, quartiles and interquartile range as estimates, so that the class can use position and spread to predict next steps for the learners and for Mr. Otieno and Mrs. Wanjiku.
Knowledge	Learners will: (1) build a cumulative frequency (less-than) table using the upper class boundaries 59.5 to 99.5 for Subject 1 and 9.5 to 69.5 for Subject 2, and check that the last cf equals $n = 20$; (2) draw an ogive by plotting (upper boundary, cf) points; (3) read the median (10th value), Q1 (5th value) and Q3 (15th value) from the ogive as estimates; (4) calculate the estimated IQR = $Q3 - Q1$ (Subject 1: 13.2; Subject 2: 24.2); (5) check the ogive median against the grouped median formula from Lesson 5 (80.75 and 33.5).
Skills	Developing and using models (NGSS SEP 2): turning a table into a graph and using the graph to find values. Analysing and interpreting data (SEP 4): reading quartiles from the ogive and comparing spread. Using mathematics and computational thinking (SEP 5): calculating cf, positions $n/4$, $n/2$, $3n/4$ and the IQR. Constructing explanations (SEP 6): using the ogive to justify a statement about the two subjects. Communication: groups of 4 present ogives on manila paper.
Attitudes	Learners will show patience and accuracy when plotting and reading graphs, respect for the evidence in the data when it differs from first impressions, and teamwork when sharing graph-drawing roles. They will be cautious about claiming reasons for the scores, using words like 'may' and 'we need more data'.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	Lessons 1 to 6 showed the raw and ordered scores, the frequency tables, the measures and the histograms and polygons. Lesson 7 adds a new tool, the ogive. It lets the class read the median and quartiles directly and measure spread with the IQR (13.2 for Subject 1, 24.2 for Subject 2). This supplies the evidence for Lesson 8, where all the summaries are used together to compare the subjects and predict next steps.
Safety Notes	Use sharp pencils and rulers carefully. Keep walkways clear when groups display manila paper. Learners share rulers and graph paper so that no one is excluded. Stay in seats while measuring on the large class graph.

B. LESSON OVERVIEW

Madam Achieng has asked the Grade 10 class at the day secondary school in Kisumu to use the data from Mr. Otieno's Subject 1 and Mrs. Wanjiku's Subject 2 to decide next steps. Learners have ordered the scores, tabulated them, calculated measures and drawn histograms and polygons. In this lesson they ask a new question: 'If we line up all 20 learners from lowest to highest, what score marks the halfway learner, the learner a quarter of the way up, and the learner three quarters of the way up?' They build cumulative frequency tables (for example, Subject 2: 13 learners scored below 39.5), plot ogives on the same axes, and read the median, Q1 and Q3 as estimates. They compare the readings with the grouped median formula from Lesson 5 and discuss why the ogive shows spread (IQR 13.2 against 24.2). Learners treat any reasons for the difference as hypotheses and avoid claiming causes.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Represent Real-World Data Using Bar Graphs, Frequency Tables and Histograms (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners look at the Subject 2 grouped frequency table from Lesson 3 and Madam Achieng's question: 'How many learners scored less than 40 in Subject 2?' They first predict individually and then in pairs. They write: (a) their estimate of the number of learners below 39.5 and (b) which score they think the middle learner (the 10th) has in each subject. Then they sketch what they think a graph of 'number of learners below each score' would look like for Subject 1 and for Subject 2, marking whether it climbs early or late. Common predictions might be: Subject 2 climbs early and Subject 1 climbs late, because Subject 1 has no scores below 67. Learners record the predictions in their books and on a class chart to revisit in the Explain Phase.	Write on the board: 'Mr. Otieno's class scored 67 to 95. Mrs. Wanjiku's class scored 11 to 60. Madam Achieng asks: how many learners are below 40 in Subject 2?' Say: 'Do not calculate yet. Use the table from Lesson 3 and make your best prediction.' WAIT TIME 10-15 seconds in silence. Then ask: 'Turn to your partner: where on the number line is the middle learner in Subject 2, and why do you think so?' Cold-call 4 pairs, including at least 2 who have not spoken in the last lesson. Ask: 'What would a graph that counts learners up to each score look like?' Accept sketches and say: 'We will test your sketches with evidence.' Do not correct predictions yet. Record them on a chart headed 'Our Predictions'.	Prediction with sketch-and-justify: learners commit to a prediction and a rough sketch before instruction, so their ideas become visible. Revisiting them later lets learners see how evidence changes their thinking, and it gives the teacher a baseline for misconceptions, such as confusing frequency with cumulative frequency.	Look for: (1) learners who write a number between 0 and 20 for learners below 40 (a sensible range) versus those who write the frequency of one class only; (2) sketches that rise from left to right (a sign of understanding 'running total'), versus sketches that look like histograms; (3) justifications that refer to the table, such as 'Subject 2 has many scores in the 10s to 30s'. Note on a class list which learners show the histogram misconception so that you can support them in the Observe Phase.
Observe Phase  VIDEO: Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating the median: data displays <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)	Part A (cumulative frequency tables, about 20 minutes): In groups of 4 learners start with the grouped table from Lesson 3 (Subject 1: 60-69: 2, 70-79: 7, 80-89: 8, 90-99: 3; Subject 2: 10-19: 4, 20-29: 4, 30-39: 5, 40-49: 3, 50-59: 2, 60-69: 2). They write the upper class boundaries and build less-than cumulative frequency (cf) columns. Subject 1: <59.5: 0; <69.5: 2; <79.5: 9; <89.5: 17; <99.5: 20. Subject 2: <9.5: 0; <19.5: 4; <29.5: 8; <39.5: 13; <49.5: 16; <59.5: 18; <69.5: 20. They check that the last cf is 20 and compare with the ordered lists. For example, in Subject 2 the scores below 39.5 are 11, 13, 15, 18, 21, 23, 27, 29, 30, 31, 32, 32, 35, which is 13. Part B (drawing	Model one row on the board: 'The class 30-39 has $f = 5$. The cf up to 39.5 is $4 + 4 + 5 = 13$. What does the 13 tell us about learners?' WAIT TIME 10-15 seconds. Cold-call 3 learners to say in words: '13 learners scored less than 39.5 in Subject 2.' Circulate while groups build tables and ask: 'Which boundary do we plot against, the lower, the upper or the midpoint? Why?' If a group uses midpoints, say: 'That is what we did for the polygon in Lesson 6. Here the cf counts everyone up to the END of the class, so what value does the end of the class have?' Prompt: 'Why does Subject 1 start at 59.5 with cf 0?' During drawing say: 'Use the same scales on both ogives so we can compare fairly.'	Data-to-graph modelling with structured roles: learners convert a table into a visual model and then extract information from the model. Roles ensure that everyone handles the data, and reading across and down from the cf axis makes the idea of 'the halfway learner' concrete. Using the same axes for both subjects supports comparison, as in the histograms of Lesson 6.	Observable indicators: (1) cf columns that increase and end at 20; (2) points plotted at upper boundaries (e.g. (39.5, 13)), not midpoints; (3) ogives that start at 59.5 and 9.5 with cf 0 and never go down; (4) reading lines drawn across at 5, 10 and 15 and then down to the score axis; (5) estimates within about 1 mark of the values: Subject 1 Q1 73.8, median 80.75, Q3 87.0; Subject 2 Q1 22.0, median 33.5, Q3 46.2; (6) learners use the word 'estimate'. Use a checklist as you circulate and mark groups that need help with boundaries or scales.

Search ARES for similar readings	ogives, about 35 minutes): Each group draws two ogives on the same axes, with score on the horizontal axis (0 to 100) and cf on the vertical axis (0 to 20). Each ogive starts at the lower boundary of the first class with cf 0 (Subject 1 at 59.5; Subject 2 at 9.5) and uses points (upper boundary, cf), joined with a smooth curve. Roles: reader of values, plotter, ruler holder, checker. Part C (reading the ogive, about 15 minutes): learners draw lines at cf = 5, 10 and 15 ($n/4$, $n/2$, $3n/4$), read the scores and record: Subject 1 Q1 about 73.8, median about 80.75, Q3 about 87.0; Subject 2 Q1 22.0, median 33.5, Q3 about 46.2. They calculate the IQRs: 13.2 and 24.2. Learners write that all are estimates because the data is grouped.	Check axes, scales and the first point. During reading ask: 'Where is the 10th learner on your graph? Draw across from 10, then down. What score do you read?' Remind learners to write 'about' or 'estimate'. Stop after 10 minutes and ask 3 groups to share their Subject 2 median reading with the class, writing the values on the board (accept 33 to 34).		
Explain Phase  VIDEO: Interquartile range (IQR) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Bar Graphs, Frequency Tables, and Histograms Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	Learners meet in their groups and compare their ogive readings with the grouped median formula from Lesson 5. They calculate Subject 1: median = $79.5 + ((10 - 9)/8) \times 10 = 80.75$, and Subject 2: median = $29.5 + ((10 - 8)/5) \times 10 = 33.5$. They compare these with their ogive readings and compare both with the exact ungrouped medians (80.5 and 31.5), noting that grouped values are estimates and differ slightly. They also verify Q1 for Subject 2 by interpolation: $19.5 + (1/4)(10) = 22.0$ and Q3: $39.5 + (2/3)(10) = 46.2$. Next they explain the shapes of the two ogives: Subject 2 rises steeply on the left (many low scores) and flattens as the scores stop at 69.5; Subject 1 stays flat at 0 until 59.5, then rises steeply between 69.5 and 89.5. Then they compare IQRs (13.2 and 24.2) and complete a	Ask: 'Your ogive says the Subject 2 median is about 33.5. The formula in Lesson 5 gave 33.5. Why would these agree?' WAIT TIME 10-15 seconds, then cold-call 3 learners. Follow up: 'The exact median from the ordered list was 31.5. Why is 33.5 not the same?' Elicit that grouped data hides the exact scores inside each class. Ask: 'Which subject has the larger IQR? What does a larger IQR tell Madam Achieng?' Cold-call 4 learners, accepting 'the middle half of learners are more spread out in Subject 2'. Challenge: 'A learner says Mrs. Wanjiku must be a weaker teacher because the scores are lower. Does the ogive prove that?' WAIT TIME 10-15 seconds. Guide: 'The data shows WHAT happened, not WHY. Rewrite the claim using the words may or could.' Have 2	Claim-Evidence-Reasoning with cross-checking: learners support statements about median and spread with two methods (ogive and formula) and the graph shape. Agreement between methods builds confidence in the model, and the 'data shows what, not why' discussion trains learners to separate evidence from hypotheses. Revisiting predictions makes learning visible.	Indicators: (1) learners show the working for $79.5 + (1/8)(10) = 80.75$ and $29.5 + (2/5)(10) = 33.5$ correctly; (2) learners say the ogive and formula values are close because both are estimates from grouped data; (3) sentence frames completed with values from 73.8 to 87.0 (width 13.2) and 22.0 to 46.2 (width 24.2); (4) statements that avoid causal claims, e.g. 'the lower scores in Subject 2 could be due to many factors, we need more data'; (5) learners explain a steep section of an ogive as 'many learners in that class'. Collect one sentence from each learner as an exit ticket.

	sentence frame: 'The middle half of Subject 1 scores lies between ___ and ___, a width of ___, while the middle half of Subject 2 lies between ___ and ___, a width of ___. This shows that ___.' They also discuss: 'Did your prediction about the climb of the graphs match?' Finally they discuss what the ogive does not tell us: it does not show why the scores differ, and reasons such as test difficulty, teaching method or revision time are hypotheses that need more data.	groups read their revised statements aloud. Return to the Predict Phase chart and ask: 'Which predictions were supported? Which should we change?' Use a different colour to mark revisions.		
Driving Question Board (DQB) Creation  VIDEO: Frequency tables & dot plots <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Frequency Tables to Organize and Display Data (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12  Search ARES for similar readings	Learners return to the Driving Question Board headed: 'How can you use the two sets of data to predict the learners' and the teachers' next steps?' Individually they write on sticky notes (a) one thing the ogive helped them to answer, for example 'Half of the learners in Subject 2 scored below about 33.5' and (b) one new question, for example 'Why is the spread of Subject 2 larger?', 'Will a second test give the same ogive?' or 'Which topics did the 16 learners below 50 find hardest?' They place notes under the headings 'Answered', 'Partly answered' and 'New questions'. As a class they link new notes to earlier notes about tables, measures and graphs and decide which will be addressed next in Lesson 8, where all the summaries will be compared.	Say: 'Take two minutes of silent thinking. What did the ogive show that the histogram and the raw list did not?' WAIT TIME 10-15 seconds before they write. Cold-call 4 learners to read their notes and ask the class: 'Does this answer the DQB, partly answer it, or raise a new question?' Move notes physically. Prompt: 'Which question needs more data, such as topic-by-topic marks or a second test?' Highlight that Madam Achieng's decision needs justified predictions. Link back: 'Key inquiry 1: how do we organise data? We now have tables, measures and three graphs. Key inquiry 2: why is it important to collect, organise and interpret data? Give one example from today.' Ask 2 learners to answer.	Driving Question Board with sorting and linking: learners categorise questions as answered, partly answered or new. This shows progress, builds a record of learning and motivates the next lesson, helping learners see that understanding grows from evidence.	Indicators: (1) notes referring to specific values (median 33.5, IQR 24.2); (2) new questions that ask for causes or more data rather than only recalling facts; (3) correct sorting of at least one note into 'Answered' with evidence; (4) learners connect their note to a previous lesson (e.g. 'the histogram showed the shape but not the middle half'). Record which learners wrote only recall questions and prompt them with 'What would you need to find out to be sure?'
Model Building Phase  VIDEO: Histograms	Learners update their group model (a one-page summary on manila paper) of the question 'How can the two sets of data predict next steps?'. They add a new section	Say: 'Add today's evidence to your model. Every prediction must point to a number from a table or graph.' Circulate and ask: 'Which value supports this prediction? What	Cumulative model revision with peer feedback: learners integrate new evidence (ogive, quartiles, IQR) into an existing model and justify predictions with data. Peer	Indicators: (1) the model includes both cf tables, both ogives, and the correct quartile and IQR estimates; (2) each prediction cites a number (e.g. 'only 4 of 20 pass', 'IQR 24.2

<i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum) Search ARES for similar videos  EXERCISE: Frequency Tables to Organize and Display Data Practice <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	called 'Position and spread' that contains: both cumulative frequency tables; a short description of each ogive (Subject 1: flat until 59.5, steep from 59.5 to 89.5; Subject 2: steep early, flattening at about 59.5 to 69.5); the estimated quartiles (Subject 1: Q1 73.8, median 80.75, Q3 87.0; Subject 2: Q1 22.0, median 33.5, Q3 46.2) and IQRs (13.2 and 24.2). They add two provisional predictions, for example: 'If the pattern holds, Mr. Otieno's class may be ready for extension work' and 'Mrs. Wanjiku and the learners may need to reteach and revise Subject 2 topics and give support to the 16 learners below 50, and collect more data to find the cause.' They label predictions as provisional. Groups swap models and give one 'two stars and a wish' comment. Finally, they complete the summary table prompt in their notebooks.	would make you change it?' WAIT TIME 10-15 seconds after each question. Cold-call 3 groups to read one prediction and its supporting evidence. Challenge any prediction that claims a cause: 'How do you know it is the teaching? What else could it be?' Ask learners to rewrite using 'may' or 'could'. Close by saying: 'Next lesson we put everything together: averages, spread, shape and pass rate to compare the two subjects.' Leave models on display.	feedback exposes gaps and tentative language keeps conclusions honest.	versus 13.2'); (3) learners use tentative language and mention more data; (4) peer feedback that names specific strengths and missing evidence. Use a 3-level rubric (beginning, developing, secure) and note groups that need help with interpreting spread.
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Did learners plot at upper boundaries and start the ogives at 59.5 and 9.5 with cf 0, or did some use midpoints as in the polygon? (2) Could learners explain what 'the 10th learner' means on the cf axis? (3) Did the ogive and formula medians (80.75 and 33.5) agree in learners' work, and did they understand why they differ from the exact 80.5 and 31.5? (4) Did learners use 'may', 'could' and 'we need more data' when discussing the score difference, rather than blaming learners or teachers? (5) Which learners need more practice with scales and reading across and down? Plan a short warm-up for Lesson 8 using one quartile reading.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the cumulative frequency of Subject 2 reaches 13 by 39.5 and 20 by 69.5, while Subject 1 stays at 0 until 59.5 and then rises to 20 by 99.5. Both ogives were drawn using (upper boundary, cf) points. From the ogives we read estimates: Subject 1 Q1 73.8, median 80.75, Q3 87.0; Subject 2 Q1 22.0, median 33.5, Q3 46.2.
What did I learn?	We learned that a cumulative frequency table adds up the frequencies to show how many learners scored below each upper boundary, and that an ogive lets us read the median (10th), Q1 (5th) and Q3 (15th) as estimates. The IQR = Q3 - Q1 is 13.2 for Subject 1 and 24.2 for Subject 2, so the middle half of Subject 2 scores is more spread out.
How does this explain the	The ogives help us to answer the driving question by showing position and spread: Subject 1 scores are high and close together,

phenomenon?	while Subject 2 scores are lower and more spread out, with a median estimate of 33.5. This supports predictions that Mr. Otieno's class may continue with extension work and that Mrs. Wanjiku and the learners may reteach and revise, give support to the learners below 50 and collect more data. The graphs show what happened but not why, so we need more data such as topic-by-topic marks and a second test before concluding.
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LESSON 8 (40 minutes (one double lesson of 80 minutes if the Model Building Phase is extended)): Interpretation of Data: Comparing Subjects and Predicting Next Steps

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners interpret tables, measures and graphs together to compare Subject 1 and Subject 2 (average, spread, shape, pass rate) and use the evidence to draw justified conclusions and predictions about the next steps for the learners, Mr. Otieno and Mrs. Wanjiku.
Knowledge	Learners will be able to: (1) compare two sets of data using the mean, median, range and interquartile range (Subject 1: mean 80.35, median 80.5, range 28; Subject 2: mean 33.85, median 31.5, range 49); (2) read the shape and spread of the two histograms/frequency polygons and the two ogives; (3) use pass counts (20 of 20 in Subject 1; 4 of 20 in Subject 2, 16 below 50, and 7 learners at 40 or above) to describe performance; (4) distinguish between what the data shows and what it cannot show (the causes).
Skills	Analysing and interpreting data (NGSS SEP 4); Using mathematics and computational thinking (SEP 5); Engaging in argument from evidence (SEP 7); Obtaining, evaluating and communicating information (SEP 8). Learners also practise critical thinking, communication and collaboration in groups of 4.
Attitudes	Learners appreciate that data can be used fairly to make decisions per subject rather than labelling learners as 'weak' or 'clever'; they show curiosity, honesty and respect for others' views when discussing results about their own class and teachers; they accept that predictions should be revised when new data arrives.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	In Lessons 1-7 the class ordered the scores, built frequency tables, calculated the measures and drew histograms, polygons and ogives. Lesson 8 brings all this evidence together. For the first time learners use it as one body of evidence to answer the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps? It prepares Lesson 9, where learners write final evidence-based recommendations and revise the final model.
Safety Notes	No hazardous materials. Handle scissors (if used to cut summary cards) carefully. Be sensitive: the scores are of one class in a day secondary school in Kisumu, so ensure no learner is blamed or laughed at. Remind learners that the data shows differences between subjects, not who is 'weak' or whose teaching is 'bad'. Keep groupwork spaces clear of bags and desks to avoid tripping.

B. LESSON OVERVIEW

Madam Achieng, the Deputy Principal of a day secondary school in Kisumu, has asked the Grade 10 class to use the data to decide next steps. The same 20 learners scored 67 to 95 in Subject 1 (Mr. Otieno) and only 11 to 60 in Subject 2 (Mrs. Wanjiku). In this lesson learners put together everything gathered so far: tables, measures (mean 80.35 vs 33.85; median 80.5 vs 31.5; range 28 vs 49; IQR 13.5 vs 22.5), graphs (histograms, polygons, ogives) and pass counts (20 vs 4). In groups of 4 they build a comparison on manila paper, practise writing claim-evidence-reasoning statements, and predict next steps while separating what the data shows from what it cannot explain (causes remain hypotheses). The lesson answers Key Inquiry 2 by showing how organised data reveals patterns that the raw lists hid.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Example: Comparing distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Comparing center and spread Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners look again at the two raw lists on the board (Subject 1: 72, 85, 90, 67, 78 ... ; Subject 2: 35, 42, 27, 58, 44 ...). Individually they write on a sticky note or in their exercise book: 'If I were Madam Achieng, the one thing I would do next for Subject 1 is ____, and for Subject 2 is ____, because ____.' Then they predict which summary measures (mean, median, range, IQR, pass count) and which graph will give the strongest evidence for their decision. They share with a partner (Think-Pair-Share) and keep their predictions for later checking.	Display the two raw lists and say: 'Last week you turned these lists into tables and graphs. Madam Achieng now wants a decision. Before we look at our summaries, write your first prediction.' Give WAIT TIME of 10-15 seconds after each question, silently counting. Ask: 'Which would help you more to decide, the raw list or the histogram? Why?' Cold-call 4 learners from different groups, including at least 2 who have not spoken yet, and record predictions on the board under two headings: 'Learners' and 'Teachers'. Say: 'Keep your prediction. We will test it against the evidence today.' Do not correct predictions; ask 'What evidence would change your mind?'	Predict-Observe-Explain with Think-Pair-Share: Writing a prediction first surfaces learners' prior ideas (such as 'Subject 2 learners are weak') and makes them accountable to evidence. Pair talk lets learners rehearse reasoning before the class discussion, so the later comparison with the data is a real test of their thinking.	Look for: (a) predictions that mention at least one measure or graph (e.g. 'the mean', 'the pass count') rather than only opinions; (b) whether learners predict a different action for each subject; (c) whether any learners blame learners' ability or the teachers without evidence. Tally these on a class list. Learners who write only opinions are prompted with 'Which number would show that?'
Observe Phase  VIDEO: Means and medians of different distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Describing distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	In groups of 4, learners receive a Data Summary Pack (prepared from Lessons 4-7) with: both ascending lists, the grouped frequency table, the two histograms/frequency polygons on the same axes, the two ogives on the same axes, and a blank comparison table. They fill in the table: Mean (80.35 / 33.85), Median (80.5 / 31.5), Mode (none / 32 and 60), Range (28 / 49), Q1 and Q3 (73.5, 87 / 22, 44.5), IQR (13.5 / 22.5), lowest and highest score (67 and 95 / 11 and 60), pass count (20 / 4), fail count (0 / 16), and learners at 40 or more (20 / 7). They then annotate the graphs: where the tallest bar is (80-89 for Subject 1; 30-39 for Subject 2), where the polygons overlap (only the 60-69 class, with 2 learners each), and where each ogive rises steepest. They note	Say: 'Your job now is to be data detectives. Look only at what the numbers and graphs show. Do not say why yet.' Circulate to every group. Ask: 'Where on the histogram can you see that all 20 learners passed Subject 1?' (WAIT TIME 10-15 seconds.) Ask: 'Which class in Subject 2 has the most learners, and how many?' Ask: 'Subject 2 has a bigger range, 49 compared with 28. What does that look like on the graph?' Prompt groups that rely on a single measure: 'The means are 80.35 and 33.85. Does the mean alone show the spread? What else tells you?' Remind: 'Grouped values are estimates, so they differ slightly from exact ones.' Cold-call 3 groups to report one observation each, and write them on the board in a 'We observe' column. Check the first group's pass count of 4	Data Discourse / Noticing and Wondering with multiple representations: Learners connect the same fact across a table, a measure and a graph (for example, 'range 49' and 'wide polygon'). Seeing one idea in three forms builds a more robust understanding and helps them see why raw lists hid patterns.	Check that the comparison table is correctly completed (means 80.35 and 33.85, medians 80.5 and 31.5, ranges 28 and 49, IQRs 13.5 and 22.5, pass counts 20 and 4). Look for learners who correctly point to the 80-89 and 30-39 classes on the graphs, correctly describe the narrow cluster against the wide spread, and use the word 'estimate' for grouped values. Note any learner who confuses the range with the IQR or the pass count with the number of learners at 40 or above (7), and address it on the spot.

	that grouped estimates (mean 80.5 and 35; median 80.75 and 33.5) differ slightly from the exact values. Each group records three observations in the form 'I notice ____ in the ____ (table/graph/measure).'	(51, 58, 60, 60) against the ascending list.		
Explain Phase  VIDEO: Mean and standard deviation versus median and IQR Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Comparing center and spread Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Each group writes three Claim-Evidence-Reasoning statements on manila paper. Example Claim: 'Subject 1 scores are higher and more tightly clustered than Subject 2 scores.' Evidence: 'mean 80.35 vs 33.85; median 80.5 vs 31.5; range 28 vs 49; IQR 13.5 vs 22.5.' Reasoning: 'A higher average with a smaller range means most learners in Subject 1 scored close to the top, while in Subject 2 scores are low and far apart.' Other required claims: (2) 'All 20 learners passed Subject 1 but only 4 passed Subject 2, so the same learners cannot all be weak; the difference is between subjects.' (3) 'In Subject 2, 7 learners scored 40 or more, and 16 are below the pass mark of 50, so some learners need remedial support.' Groups then write predicted next steps: for learners, Mr. Otieno and Mrs. Wanjiku. They also write one 'We cannot yet say' statement, for example 'We cannot say why Subject 2 scores were low; the test may have been harder, the topics may not have been revised, or the teaching method may differ. We need more data.' Groups display their work in a gallery walk, and each learner adds one star and one question to another group's paper.	Model one CER on the board with Subject 1 and say: 'A good claim is checked by numbers. Which number proves it? How does the number prove it?' Insist: 'Use the words may, could, we need more data when you talk about reasons.' Ask: 'Mrs. Wanjiku's class has a lower mean. Does that prove she is a poor teacher?' (WAIT TIME 10-15 seconds.) Guide learners to see that the data cannot tell us. Ask: 'Could the test have been harder? What data would show that?' (Topic-by-topic marks, a second test.) Cold-call 5 learners during the gallery walk to explain another group's claim in their own words. Challenge outliers: 'Subject 1's highest score is 95, Subject 2's is 60. Does a high maximum mean the subject is easier? What else do we check?' Use Kenyan comparisons if needed: 'If one farmer in Kericho picks tea quickly and another is slower, can we say why from the kilograms alone?'	Claim-Evidence-Reasoning (argument from evidence) with a Gallery Walk: CER forces learners to connect each claim to specific numbers and to explain how the numbers support the claim. Distinguishing 'what the data shows' from 'what we cannot yet say' builds scientific and statistical honesty. The gallery walk exposes learners to alternative reasoning and prompts peer critique.	Use a checklist for each group: Does each claim cite at least two numbers? Is the reasoning linked to spread and average, not only to the average? Does the group avoid claiming causes as facts? Do they state 16 below 50 and 7 at 40 or above correctly? Successful indicators: learners use 'may' or 'could' when giving reasons. If a group writes 'Mrs. Wanjiku teaches badly', ask them to rewrite the statement as a hypothesis that could be tested.
Driving Question Board (DQB)	Learners return to the Driving Question Board with the question 'How can you use the two sets of	Say: 'Look at the DQB. Which of our questions can we now answer with numbers? Which cannot be	Driving Question Board with question sorting: Sorting questions into 'answered by evidence' and	Observe whether learners can support 'Answered' with a specific piece of evidence (such as pass

Creation  VIDEO: Example: Comparing distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Describing distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	data to predict the learners' and the teachers' next steps?' They move sticky notes: questions answered by the evidence (for example 'Did the same learners pass both subjects?' - answered: all 20 passed Subject 1, 4 passed Subject 2) go to an 'Answered' column with a short evidence note. New questions are added, such as 'Was the Subject 2 test harder than the Subject 1 test?', 'Which Subject 2 topics lost the most marks?', 'Will a second test show the same pattern?', 'Would the same pattern appear in other classes?' Each learner writes at least one new question and one answer with evidence, referring to the Kisumu class data.	answered with the data we have?' Give WAIT TIME of 10-15 seconds. Cold-call 4 learners to move a question to 'Answered' and justify it with a number. Ask: 'What new question has today's comparison raised?' Highlight questions about causes and say: 'Those are questions for more data. In Lesson 9 we will decide which data to collect and write our recommendations.' Ensure that each group adds at least one question. Point out the link to Key Inquiry 2: 'Which question on the board shows why collecting and interpreting data matters?'	'needs more data' makes progress visible and turns gaps in the evidence into purposeful next investigations. It keeps the learners' own questions at the centre of the unit.	count 20 vs 4, or range 28 vs 49). Check that new questions are testable (for example, 'topic-by-topic marks' rather than 'Is Mrs. Wanjiku good?'). Note learners who add questions about causes and praise them as good scientific thinking.
Model Building Phase  VIDEO: Means and medians of different distributions Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Calculating the mean <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Learners revise their class model, a flow diagram on manila paper titled 'From Raw Scores to Decisions': Raw lists -> Ascending order -> Frequency tables -> Mean, median, mode, range, quartiles -> Histograms, polygons and ogives -> Comparison and interpretation -> Predictions and next steps. In this lesson they add the Interpretation stage with the boxes: 'Compare average (80.35 vs 33.85)', 'Compare spread (range 28 vs 49, IQR 13.5 vs 22.5)', 'Compare pass rate (20 vs 4)', and 'What we cannot say yet: causes'. They then draft predictions: Mr. Otieno's class: continue and give extension or enrichment work; Mrs. Wanjiku and the learners: reteach and revise the Subject 2 topics, remedial support for the 16 learners below 50, and collect more data (topic-by-topic marks, a second test). Learners also compare these with their original	Say: 'Your model should now show how the data led you to a decision. Where does today's work fit?' Ask groups to point to the arrow where interpretation happens. Ask: 'If the Subject 2 mean rises to 50 in the second test, which parts of your model would change?' (WAIT TIME 10-15 seconds.) Cold-call 3 groups to read their predictions and justify each with one number. Press: 'Is this a prediction or a fact? What word shows that?' Check that learners say 'estimate' for grouped measures. Collect exit tickets. Tell the class: 'In Lesson 9 we write the final recommendations for Madam Achieng and revise this model for the last time.'	Cumulative Model Revision with Prediction Reflection: Adding the interpretation stage to the evolving model shows how each lesson's skill (ordering, tabulating, summarising, graphing) contributes to decisions. Comparing the new predictions with the Predict Phase ideas develops metacognition and shows that evidence can change thinking.	Check that the revised model includes the interpretation stage and the 'cannot say yet' box. In the exit ticket, look for: a correct number from a table (such as the pass count 20 vs 4), a correct measure (such as the mean 80.35 vs 33.85 or range 28 vs 49), a reference to a graph (histogram, polygon or ogive), and a conclusion for each subject. Proficient responses state that decisions should be made per subject and that more data is needed to find causes.

	predictions from the Predict Phase and write one sentence: 'My first prediction changed/stayed the same because ____.' Finally, each learner completes an exit ticket: 'Using one table, one measure and one graph, explain why Madam Achieng should make a separate decision for each subject.'			
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D. TEACHER REFLECTION

After the lesson, reflect: (1) Which groups used both an average and a spread measure in their claims, and which relied only on the mean? (2) Did any learners write causes as facts (for example 'Mrs. Wanjiku is not teaching well')? How did I respond, and how can I reinforce the idea of hypotheses in Lesson 9? (3) Could learners read the ogives and histograms accurately and distinguish the exact values (80.35, 33.85) from the grouped estimates (80.5, 35)? (4) Did learners confuse the 16 learners below 50 with the 7 learners at 40 or above? (5) Were cold-calls distributed fairly, and did every learner speak at least once? (6) Which DQB questions need to be addressed in Lesson 9, and what extra data (topic-by-topic marks, a second test) will the class suggest? Record two changes to make next time.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that Subject 1 scores are high and tightly clustered (mean 80.35, median 80.5, range 28, all 20 learners pass) while Subject 2 scores are low and widely spread (mean 33.85, median 31.5, range 49, only 4 of 20 pass, 16 below 50, 7 at 40 or more). The histograms, polygons and ogives show the same pattern: the Subject 1 graphs are narrow and to the right, and the Subject 2 graphs are wide and to the left.
What did I learn?	We learned that tables, measures and graphs must be read together to compare two sets of data using average, spread, shape and pass rate. We learned that the grouped values are estimates, and that the data shows differences between subjects but cannot tell us why the scores differ.
How does this explain the phenomenon?	We explain that the same 20 learners cannot all be weak, because they all passed Subject 1 but most failed Subject 2, so decisions must be made per subject. We predict that Mr. Otieno's class should continue with extension or enrichment work, and that Mrs. Wanjiku and the learners should reteach and revise the Subject 2 topics, give remedial support to the 16 learners below 50, and collect more data (such as topic-by-topic marks and a second test) before concluding why the scores differ.

LESSON 9 (40 minutes (1 double lesson of 80 minutes if available; core activities fit in 40 minutes)): Why Collecting, Organising and Interpreting Data Leads to Effective Decisions

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To bring together everything learned in Lessons 1-8 to write evidence-based recommendations that answer the driving question, and to show why collecting, organising and interpreting data supports effective decisions.
Knowledge	Learners will explain, using the Key Comparison, that Subject 1 scores are high and tightly clustered (mean 80.35, median 80.5, range 28, all 20 pass) while Subject 2 scores are low and widely spread (mean 33.85, median 31.5, range 49, 4 of 20 pass), and that organised data shows patterns that the raw lists hid.
Skills	Learners will interpret tables, measures and graphs together; justify recommendations using numerical evidence; separate what the data shows from what it cannot show (causes); compare their Lesson 1 model with the final model; and communicate conclusions on manila paper (NGSS SEP: Engaging in Argument from Evidence; Developing and Using Models; Obtaining, Evaluating and Communicating Information).
Attitudes	Learners will value evidence over opinion, show fairness to learners and teachers by not blaming without data, and collaborate respectfully in groups of 4.
Key Inquiry Question	1. How do we organise data using frequency distribution tables? 2. Why is it important to collect, organise and interpret data in making decisions?
Purpose in Storyline	This is the final lesson. In Lesson 1 learners only had raw and ordered lists and could say who passed or failed. Lessons 2-3 organised the data into frequency tables, Lessons 4-5 summarised it with mean, median and mode, Lessons 6-7 graphed it (histograms, polygons, ogives, quartiles) and Lesson 8 compared the subjects. Lesson 9 turns that evidence into the final explanation and recommendations that answer the driving question: How can you use the two sets of data to predict the learners' and the teachers' next steps?
Safety Notes	No hazardous materials. Remind learners to handle manila paper and markers sensibly, keep walkways clear during the gallery walk, and speak respectfully about learners and teachers: we discuss data, not people. Be sensitive to learners who may personally relate to low scores; frame results as information about the subject test, not about any learner's ability.

B. LESSON OVERVIEW

Madam Achieng, the Deputy Principal of the day secondary school in Kisumu, has asked the Grade 10 class for a final report on the two sets of scores of the same 20 learners. Learners revisit the Driving Question Board, collect the evidence gathered across Lessons 1-8, and use it to write recommendations for Mr. Otieno (Subject 1), Mrs. Wanjiku (Subject 2), the learners, and the 16 learners below 50 in Subject 2. They compare their Lesson 1 model (raw and ordered lists, pass/fail only) with a final model (tables, measures, graphs and interpretation), complete the DQB, and write a Final Explanation of why collecting, organising and interpreting data leads to effective decisions. Throughout, reasons for the score difference are treated as hypotheses to investigate (test difficulty, teaching method, revision time are unknown), not facts.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Interpreting a histogram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Represent Real-World Data Using Bar Graphs, Frequency Tables and Histograms (Flexbook) Source: CK-12  Search ARES for similar readings	Learners recall the phenomenon: the same 20 learners scored 67 to 95 in Subject 1 and 11 to 60 in Subject 2. Individually, then in pairs, they write a 2-3 sentence prediction on a sticky note or in their notebooks: 'If Madam Achieng only had the raw lists, what would she decide? If she has all our tables, measures and graphs, what will she decide, and how sure can she be?' They also predict which of these decisions are safe to make now and which need more data (for example, 'Are the learners weak?' or 'Is the Subject 2 test too hard?').	Display the two raw lists on the board next to a blank space. Say: 'Madam Achieng is waiting for our report. Before we write it, take two minutes silently. What would she decide from the raw lists alone? What could she decide now that we have organised the data?' WAIT TIME: 15 seconds after the question before anyone speaks. Circulate and note 3 different predictions. Cold-call 4 learners (not volunteers) to share, including at least one who predicted 'the learners are weak'. Record predictions on the board under the heading 'Predictions Today'. Do not correct yet; say: 'We will test these predictions against our evidence.'	Predict-Observe-Explain (prediction before evidence): Committing to a prediction makes learners aware of their starting ideas, so when the evidence is revisited they can see how much their thinking has changed since Lesson 1, and any misconception (for example 'learners are weak') becomes visible and testable.	Look for: (a) learners who predict a single decision for all 20 learners across both subjects (misconception: learners are simply 'weak' or 'strong'); (b) learners who name at least one measure or graph they would use; (c) learners who write 'we need to know why' or similar, showing awareness that causes are unknown. Tally the three types of responses on the board to guide later questioning.
Observe Phase  VIDEO: Mean and standard deviation versus median and IQR Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Represent Real-World Data Using Bar Graphs, Frequency Tables and Histograms (Flexbook) Source: CK-12  Search ARES for similar readings	In groups of 4, learners lay out the evidence 'portfolio' they have built over Lessons 1-8: the ordered lists and pass/fail counts (Lesson 1), the frequency tables (Lessons 2-3), the means, medians and modes (Lessons 4-5), the histograms and frequency polygons (Lesson 6) and the ogives and quartiles (Lesson 7). They fill in an Evidence Summary Table on manila paper with these rows for Subject 1 and Subject 2: mean (80.35 / 33.85), median (80.5 / 31.5), mode (none / 32 and 60), range (28 / 49), IQR (13.5 / 22.5), lowest and highest score (67 and 95 / 11 and 60), number passing (20 / 4), modal class (80-89 / 30-39). They also note the 7 learners who scored 40 or more in Subject 2 and that 16 of 20 are below the pass mark of 50. Each group checks that every figure is traceable to a table or graph, and	Say: 'Open your portfolios. Your job is to gather the evidence, not to decide yet. Every number you write in your Evidence Summary Table must have a source: which table or graph did it come from?' Circulate with a checklist; ask each group: 'Show me where the 31.5 comes from.' and 'Where on your histogram do you see the spread of 49?' WAIT TIME: 10 seconds after each question before prompting. Cold-call 3 groups to read one row each of their table aloud while the class checks against the Key Comparison on the board. Where a group's figure differs (for example grouped estimate 80.5 vs exact 80.35), prompt: 'Why might these two means differ slightly?' (grouped values are estimates using midpoints). Keep the anchor numbers exactly as given; do not accept invented values.	Evidence Table with Source Tracing (Obtaining and Evaluating Information): Requiring a source for each value builds the habit of tying claims to data, and placing both subjects side by side makes the pattern of higher and tighter versus lower and wider visible at a glance, which the raw lists hid.	Observable indicators: every row of the table is filled correctly with values matching the established data; learners can point to the correct table or graph for each value; learners distinguish exact values (80.35, 33.85) from grouped estimates (80.5, 35); learners state that 20 of 20 pass in Subject 1 and 4 of 20 pass in Subject 2. Note groups that mix up mean and median or misread the IQR for targeted support.

	agrees on the shape of each histogram (Subject 1 clustered between 60 and 99, Subject 2 spread from 10 to 69 with most below 50).			
Explain Phase  VIDEO: Interpreting a histogram Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Mean, median, and mode <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	Groups write evidence-based recommendations for four audiences using a 'Claim-Evidence-Reasoning-Limit' frame: (1) Mr. Otieno and the Subject 1 class: continue and give extension or enrichment work, because all 20 learners passed with scores tightly clustered (mean 80.35, median 80.5, range 28); (2) Mrs. Wanjiku and the learners: reteach and revise the Subject 2 topics, because the mean is 33.85, the median is 31.5 and only 4 of 20 passed; (3) the 16 learners below 50 in Subject 2: remedial support, with the 7 learners at 40 or above as those closest to the pass mark; (4) Madam Achieng: collect more data, such as topic-by-topic marks and a second test, before concluding why Subject 2 scores are low. For each recommendation, learners write the Claim, the numerical Evidence, the Reasoning (why this evidence supports the claim) and a Limit (what the data cannot tell us, for example 'we do not know if the Subject 2 test was harder, how it was taught, or how long learners revised'). They also write one sentence answering: 'Can the same learners all be weak?' (No: the same 20 learners did well in Subject 1, so the difference is between subjects and decisions must be made per subject.)	Say: 'Madam Achieng does not want opinions. She wants recommendations she can trust. For each recommendation, give a claim, the numbers that support it, your reasoning, and one thing the data cannot tell us.' Model one example on the board for Mr. Otieno, then let groups write the other three. WAIT TIME: 15 seconds of silent thinking before group writing begins. Circulate and ask: 'Which number convinces you most? Why that one and not the mean alone?' and 'Is that a fact from the data or a hypothesis we still need to test?' Challenge any statement such as 'Mrs. Wanjiku taught badly' or 'the test was too hard' with: 'What data shows that? What data would we need?' Cold-call 4 groups (one per audience) to read their recommendations; ask 2 other groups to give a thumbs-up or thumbs-down on whether the evidence supports the claim and why. Use 'may', 'could' and 'we need more data' in your own language.	Claim-Evidence-Reasoning-Limit (Engaging in Argument from Evidence): Linking each recommendation to specific numbers and to an honest limit teaches learners that effective decisions rest on organised data and that unexplained causes are hypotheses, not conclusions. Peer critique strengthens reasoning and corrects overreach.	Observable indicators: each recommendation cites at least two correct figures from the Key Comparison; learners include a limit that names unknown causes (test difficulty, teaching method, revision time) without asserting any as true; learners decide per subject rather than labelling learners 'weak'; learners recommend collecting more data (second test, topic-by-topic marks). Use a 3-level checklist (claim only / claim with numbers / claim, numbers and limit) and record each group's level.
Driving Question Board (DQB)	Learners revisit the Driving Question Board and the question: 'How can you use the two sets of	Say: 'Our board has grown since Lesson 1. Let us see what we can now answer.' Hold up three	Driving Question Board sorting (tracking growing understanding): Moving questions between	Observable indicators: learners justify card moves with specific data (for example a measure or

Creation  VIDEO: Mean and standard deviation versus median and IQR Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  EXERCISE: Interquartile range (IQR) <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar readings	data to predict the learners' and the teachers' next steps?' In groups they sort the DQB questions generated across the unit (for example 'Why are the raw lists hard to read?', 'How can we show spread?', 'Which average should we trust?', 'Why did the same learners score differently?') into three columns: Answered by our data, Partly answered, Needs more data. Each group moves two question cards and justifies the move with a piece of evidence (for example 'How can we show spread?' is answered by range 28 versus 49, IQR 13.5 versus 22.5, and the histograms). Learners add at least one new question to the 'Needs more data' column (for example 'Which topics in Subject 2 did the 16 learners find hardest?', 'Will a second test give similar results?', 'Would the same pattern appear in Subject 1 if the test were harder?'). Finally, learners write a one-sentence answer to the driving question on a card and pin it at the top of the DQB.	question cards one at a time and ask: 'Which column does this belong in, and what is your evidence?' WAIT TIME: 10 seconds before taking responses. Cold-call 5 different learners to move a card and justify the move; ask a follow-up to a second learner: 'Do you agree? What evidence supports or challenges that?' Highlight that the question 'Why did the same learners score differently?' stays in 'Needs more data' because the data does not tell us the cause. Praise questions that identify new data to collect. Photograph the final DQB for the class record.	columns with evidence makes progress visible, shows learners which questions data can answer and which require further collection, and links the two KICD key inquiry questions to the class's own questions.	graph); they correctly leave the 'why' question under 'Needs more data'; they generate at least one new, testable question about data to collect; their one-sentence answer to the driving question names both subjects and a next step. Record learners who cannot justify a move for follow-up.
Model Building Phase  VIDEO: Statistics intro: Mean, median, & mode <i>Related topic: not an exact match for this lesson.</i> Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Stem-and-Leaf Plots, Mean,	Learners take out their Lesson 1 model (a sketch or short note showing the raw and ordered lists, who passed or failed, and the questions about patterns, spread and action needed). On a fresh half of manila paper they build a Final Model as a flow diagram: Collect (the 20 scores in each subject) → Organise (ascending order, frequency tables with class width 10, cumulative frequency) → Summarise (mean, median, mode, range, quartiles) → Graph (histogram, frequency polygon, ogive) → Interpret (Subject 1 high and clustered; Subject 2 low and spread out; same learners, so	Say: 'Place your Lesson 1 model next to your Final Model. What can you see now that you could not see then?' WAIT TIME: 15 seconds of silent thinking, then 2 minutes of pair talk. Provide sentence starters on the board: 'In Lesson 1 we could only say... Now we can say... because...' and 'Organising the data helped us see... which the raw lists hid.' Circulate and press for specifics: 'Which graph shows the spread best?' and 'Where does your model show that we do not yet know the cause?' Cold-call 4 learners to read their Final Explanation sentences; invite 2	Model Comparison (Developing and Using Models): Placing the Lesson 1 model beside the final one lets learners see their own conceptual growth, and writing a Final Explanation requires them to connect organising, summarising, graphing and interpreting to decision-making. The gallery walk with star and wish feedback strengthens communication of evidence.	Observable indicators: the Final Model includes all stages (collect, organise, summarise, graph, interpret, decide, collect more data); the comparison table identifies at least two things the raw lists hid (spread, clustering, differences between subjects, pass rate); the Final Explanation uses at least three correct values (for example 80.35, 33.85, 28, 49, 20 of 20, 4 of 20) and states that causes are hypotheses needing more data; learners present the recommendations per subject. Use a 4-point rubric (model completeness, use of evidence, recognition of limits, clarity of

Median and Mode (Flexbook) <i>Related topic: not an exact match for this lesson.</i> Source: CK-12 Search ARES for similar readings	differences are between subjects) → Decide (recommendations per subject) → Collect more data (second test, topic-by-topic marks). Beside the model, learners complete a comparison table: 'Lesson 1 model: what we could see / could not see' versus 'Final model: what we can see now'. They then write the Final Explanation in 5-6 sentences answering the second key inquiry question: 'Why is it important to collect, organise and interpret data in making decisions?' using at least three numbers from the Key Comparison. Groups carry out a short gallery walk: each group leaves one member to explain while the others visit, then they swap roles and give each other one star (strong evidence) and one wish (a missing limit or number).	others to add or challenge a point. Close by saying: 'So, how can the two sets of data be used to predict the learners' and the teachers' next steps?' and take responses from 3 more learners. Collect the Final Explanations and models for assessment.		communication).
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D. TEACHER REFLECTION

Which groups moved from 'the learners are weak' to 'the decision must be made per subject' and what evidence convinced them? Did learners use the words 'may', 'could' and 'we need more data' when discussing the reasons for the Subject 2 results, or did some assert causes such as a hard test or poor teaching? Which part of the evidence (tables, measures or graphs) did learners use most confidently, and which needs revisiting? Were the grouped estimates (80.5 and 35) clearly distinguished from the exact values (80.35 and 33.85)? How well did the recommendations for the 16 learners below 50 balance support with respect? What would I change about how I used wait time and cold-calling so that quieter learners contributed to the Final Explanation?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Same 20 learners: Subject 1 scores 67 to 95 (mean 80.35, median 80.5, range 28, all 20 pass); Subject 2 scores 11 to 60 (mean 33.85, median 31.5, range 49, only 4 pass, 16 below 50, 7 at 40 or above). Tables, measures and graphs showed these patterns clearly, while the raw lists hid them.
What did I learn?	Collecting, organising (ordering, frequency tables) and interpreting (measures and graphs) data shows centre, spread and pass rates, so decisions can be based on evidence. Because the same learners did well in one subject and poorly in the other, decisions must be made per subject. The data cannot tell us why, so causes are hypotheses to be investigated.
How does this explain the phenomenon?	Mr. Otieno's class should continue with extension or enrichment work; Mrs. Wanjiku and the learners should reteach and revise the Subject 2 topics and run remedial support for the 16 learners below 50; and more data (topic-by-topic marks, a second test) should be collected to find the cause before concluding. Organised and interpreted data lets us predict and justify next steps for learners

	and teachers.
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DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.